Local Gambling Preference and Mortgage Misrepresentation

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Abstract

During the housing boom preceding the Great Recession, higher household leverage was sometimes achieved through mortgage misrepresentation. This paper explores a large sample of mortgages originated between 2005 and 2007, focusing on the misreporting of second liens. We find that second-lien misreporting is more severe in areas with stronger gambling preferences, controlling for other local demographic characteristics and detailed loan and borrower characteristics. In contrast, local demographics related to household risk preference, ethics, culture, and financial literacy are insignificant in explaining second-lien misreporting. We also find that second-lien misreporting is more severe when borrowers face higher incentives to misreport. The relationship between second lien misreporting and local gambling preference is robust in subsamples of borrower income, credit scores, and local housing price dynamics. Employing the FICO 620 threshold, we find little evidence that lenders' practices explain the relationship. Finally, we show that the negative performance consequences of second-lien misreporting are amplified in high-gambling-preference areas. We interpret our findings as evidence that borrowers' gambling preferences contribute to the excessive household leverage before the financial crisis.

Keywords: Excessive leverage, Local gambling preference, Second-lien misrepresentation

JEL Codes: G4, G5, R2, R3

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1 Introduction

Household financial decisions have significant effects on the aggregate market (Campbell, 2006). A prominent example is the rapid expansion of U.S. household leverage, which played a central role in the 2008 financial crisis. From the expansion of credit supply to overly optimistic expectations about house prices, researchers find that households across all income groups took on excessive leverage in the mortgage market (Mian and Sufi, 2009; Adelino et al., 2016)[1]. While much of this literature debates the underlying drivers, distorted beliefs versus misaligned incentives, relatively little attention has been paid to heterogeneity across households in such an aggressive borrowing environment. This paper provides empirical insights into which borrowers were more likely to take on excessive leverage.

In the years preceding the crisis, a striking example of excessive leverage was the misrepresentation of second liens, in which the existence of a simultaneous second lien was misreported at the time of first-lien origination (Piskorski et al., 2015; Griffin and Maturana, 2016). A standard first lien mortgage already places a household in a leveraged position, and the addition of a second lien further increases leverage by reducing homeowner equity. Yet loans with such high leverage were often difficult to sustain, especially when they combined high second lien loan to value ratios, risky first mortgages, or low documentation features. Yavas and Zhu (2024) show that the concealment of second liens was concentrated in these fragile loans, which were perceived as harder to sell in secondary markets. This finding suggests that, under truthful reporting, such loans would have been less attractive for lenders to approve because of their heightened risk and limited marketability. By hiding the second lien, borrowers were able to make their applications appear less risky and more likely to be approved. In this way, misreporting allowed households to obtain leverage that would have been difficult to achieve under truthful reporting. Therefore, we focus on the second lien misrepresentation in this study.

[1] Mian and Sufi (2009) show that subprime borrowers gained excessive access to mortgage credit that became disconnected from income growth, while Adelino et al. (2016) argue that middle- and high-income borrowers also increased leverage substantially and contributed to the rise in delinquencies during the crisis.

We start by investigating how local characteristics affect the occurrence of second-lien misrepresentation. Controlling for other local demographic characteristics and detailed loan and borrower characteristics, we find that second-lien misrepresentation is more prevalent in areas with higher Catholic-to-Protestant ratios (CPRATIO), which we interpret as a measure of local gambling preference following Kumar et al. (2011). In contrast, local demographics, such as religiosity and education, which related to household risk preference, ethics, culture, and financial literacy, are not robust in explaining second-lien misreporting.

Our interpretation is that local gambling preference affects borrowers' financial decisions, making them more willing to engage in opportunities with right-skewed payoffs. From households' perspective, second-lien misrepresentation resembles such an opportunity. The upside is obtaining better loan terms or gaining access to homeownership, which is emotionally tied to the American Dream and financially offers the possibility of wealth accumulation[2]. If house prices rise strongly, the gains can be very large, while if prices rise only moderately or stagnate, the returns are smaller and may even turn negative. At the time, however, most households and lenders believed that house prices would continue to rise, so the expectation of flat or negative returns was uncommon. The downside is limited to the costs of application, the effort of switching lenders, and the relatively weak legal penalties before the crisis[3]. This combination of potentially very large gains with relatively modest losses produces a right-skewed payoff structure that is attractive to gambling-prone borrowers.

The borrower's gambling preference interpretation implies that the effect should be stronger when incentives to misreport are greater. Christensen et al. (2018) show that local gambling attitudes increase corporate misreporting, and that the effect is amplified when incentives are larger. Following this logic, we examine subsamples that differentiate borrower incentives using property occupancy and loan purpose. The incentive is stronger for

[2]See Clinton (1995); Bush (2003) for how homeownership is part of the American Dream. See Bostic and Lee (2008); Goodman and Mayer (2018) for how, with appropriate mortgage terms, homeownership brings large financial returns and helps households accumulate wealth.

[3]Loan denial was common, but the cost of loan application and the cost of switching lenders were relatively low. Legal penalties (Henning, 2009; Federal Housing Finance Agency, 2023) were usually limited pre-crisis when law enforcement focused on fraud for profit rather than fraud for housing (FBI, 2006, 2007).

primary residences, where approval secures stable housing, than for non-primary residences, which are often discretionary or investment properties. The incentive is also stronger for purchase loans, where approval is necessary to obtain the property, than for refinancing loans, where the borrower already owns the home. Consistent with this reasoning, we find that the positive effect of local gambling preference on misrepresentation is more pronounced in these high-incentive subsamples.

We also examine subsamples defined by borrower income, credit scores, and local housing price dynamics. These dimensions are connected to debates about the origins of the financial crisis (Adelino et al., 2018), but they also capture variation in borrowers' incentives to misreport. We find that the effect of local gambling preference is present across subsamples and is stronger where borrower incentives are higher. These results are consistent with the interpretation that gambling preference shapes misrepresentation behavior and complement existing explanations of the crisis.

One concern with the interpretation is that local gambling preference may capture not only borrower behavior but also lender practices, since lenders were likely aware of the existence of second-lien misrepresentation (Piskorski et al., 2015). To address this, we use the Keys et al. (2010) setting, where the ease of securitization for low-documentation loans with FICO scores at or above 620 provides a shock to lenders in screening and potentially facilitating misrepresentation(Griffin and Maturana, 2016). We implement a regression discontinuity design (RDD) to estimate jumps in the number of loans and default rates, and apply a difference-in-discontinuities (diff-in-disc) method to compare these across high and low gambling preference areas. In the pooled sample, lenders appear to facilitate simultaneous seconds in general, but not specifically misreported seconds, as the jump ratios are smaller for misreported than for correctly reported seconds. Moreover, the default-rate discontinuity is insignificant for misreported seconds, suggesting that lax screening was not more pronounced for such loans. Importantly, when comparing high and low gambling preference areas, jumps in both the number of loans and default rates for misreported seconds are smaller in high gambling preference areas, implying that lenders in these areas do not play a larger role in facilitating misrepresentation. Taken together, these results provide little evidence that lender practices explain the relationship between gambling preference and misrepresentation.

Finally, given this gambling interpretation, we examine the outcomes of gambling preference when it is linked to mortgage misrepresentation, as misrepresentation is typically associated with poor loan performance (Piskorski et al., 2015; Griffin and Maturana, 2016). We test whether gambling preference worsens loan performance through misrepresentation. We find that loans with second-lien misrepresentation perform worse on average, and the effect is more pronounced in high gambling preference areas, indicating that gambling-motivated misrepresentation leads to poorer loan performance. A back-of-the-envelope calculation suggests that roughly \$0.8 billion of mortgage balances were affected through this channel, underscoring the economic importance of gambling attitudes in the mortgage market.

To reinforce the findings and interpretation, we conduct a series of robustness checks. First, to address functional form concerns, we re-estimate the models using different specifications and find consistent results. The probit models account for the binary nature of the dependent variable, while the causal forest approach of Wager and Athey (2018) further strengthens causal inference. Second, to reinforce the gambling preference interpretation, we examine alternative measures of gambling propensity and find similar results. Third, to validate the discussion of the lender channel, we confirm that the RDD and difference in discontinuities are valid and that the results are not sensitive to specification choices. Fourth, to address the concern that the use of jumps in number of loans is an aggregate measure that does not incorporate the loan-level controls of the baseline regressions, which may create inconsistency across specifications rather than rejecting the lender channel, we instead use the misrepresentation rate as the outcome and re-estimate the models with the full set of controls and fixed effects. The results remain consistent, further supporting our interpretation. Fifth, to examine whether the adverse outcomes of gambling-related misrepresentation differ across contexts, we perform subsample analyses based on local income, borrower credit scores, and housing price appreciation. Across all these groups, misrepresentation combined with high gambling preference produces worse loan outcomes, underscoring the robustness of our interpretation. Finally, to address outcome measurement, we examine loan performance using alternative definitions of default and apply causal forest analysis, both of which confirm that the results persist.

Our paper contributes to several strands of literature. First, it emphasizes the importance of heterogeneity in borrower characteristics, especially behavioral traits, in understanding household decisions to take on excessive leverage in the years leading up to the financial crisis. Prior research highlights how credit supply expansion and distorted house price beliefs contributed to households taking on excessive leverage (e.g., Mian and Sufi, 2009; Adelino et al., 2018; Justiniano et al., 2019). While these studies focus on aggregate forces that broadly shaped household borrowing, my paper highlights behavioral heterogeneity by showing that differences in local gambling preferences help explain why some households were more willing to take on excessive leverage than others.

Second, our paper complements the literature by discussing where in the credit supply chain (from borrowers to MBS underwriters) second-lien misrepresentations took place. By studying privately securitized loans, Griffin and Maturana (2016) point out that lenders were likely aware of the misrepresentation, attributing such misreporting more to lenders than to underwriters. Conversely, Piskorski et al. (2015) show that underwriters could easily uncover the misrepresentation and also play an important role in this issue. Using both portfolio loans and securitized loans, Yavas and Zhu (2024) support the argument that second-lien misrepresentation occurs in the early stages of intermediation by lenders, while underwriters have limited but significant effects on reducing the occurrence of misrepresentation through screening. Our paper examines this issue from the borrower's perspective, showing that borrowers also significantly contribute to second-lien misrepresentation.

Third, our paper contributes to the literature on mortgage fraud by proposing a preference-based explanation. A substantial body of literature documents the prevalence of mortgage fraud and its economic impact (Elul et al., 2010; Piskorski et al., 2015; Ambrose et al., 2016; Griffin and Maturana, 2016; Mian and Suf, 2017; Kruger and Maturana, 2021). While previous studies have considered associated loan characteristics and household features, the role of preference in the decision-making process remains unclear. From a cultural perspective, Conklin et al. (2022) provide empirical evidence that religiosity helps constrain fraudulent activity but do not distinguish between ethics and risk channels. Bypassing the issue of differentiating ethics and risk preference, we show that gambling preference[4] influences mortgage applications.

Fourth, our paper builds on the empirical literature linking gambling preferences with investment decisions. Prior research has examined the effect of gambling preference on the stock market (Barberis and Huang, 2008; Kumar, 2009; Kumar et al., 2011; Barberis et al., 2016, 2021), bond market (Benartzi and Thaler, 1995), and option market (Baele et al., 2019). Additionally, literature explores its impact on corporate policy decisions (Kumar et al., 2011; Chen et al., 2014) and fund strategies (Shu et al., 2012). To the best of our knowledge, we are the first to apply the concept of gambling preference to the mortgage market in the context of household investment decision-making. This is important because existing literature on behavioral biases in the housing market predominantly centers on pricing dynamics and speculative bubbles from the investment perspective (Lux, 1995; Genesove and Mayer, 2001; Farlow, 2004; Shiller, 2005; Farlow, 2013), whereas relatively little attention has been paid to behavioral influences on mortgage choices, which is the financing side of housing decisions.

The rest of the paper is organized as follows. Section 2 describes the data and variables. In Section 3, we examine the effect of local demographic characteristics on second-lien misrepresentation. In Section 4, we interpret the effect of gambling preference and assess whether it aligns with borrower behavior or lender practice. Section 5 investigates the outcomes of gambling preference-related misrepresentation. Robustness checks are conducted

[4]See Kumar (2009), Kumar et al. (2011), Kumar et al. (2016) for how to differentiate gambling preference from ethics.

in Section 6, and conclusions are presented in Section 7.

2 Data and Summary Statistics

Our sample comprises three main groups of data: loan-related records, religiosity data, and demographic information, covering the period from 2005 to 2007. These data are sourced from various datasets. Loan-related records include loan-level mortgage data from BlackBox Logic (now part of Moody's) and borrower-level credit report information from Equifax. Religiosity data, which provides county-level information on prevalent religious adherence, is obtained from the American Religion Data Archive (ARDA). Demographic information at the county and zipcode levels, such as age and income, is sourced from the U.S. Census Bureau. We also collect crimes and voter registration data from National Neighborhood Data Archive (NaNDA) and the Uniform Crime Reporting (UCR) Program Data available through the ICPSR website. Additionally, we collect house price data from the Federal Housing Finance Agency (FHFA) and unemployment data from U.S. Bureau of Labor Statistics.

2.1 Second-lien Misrepresentation Measure

Second-lien misrepresentation occurs when a first-lien loan backed by property is reported as having no associated higher lien but is actually financed with a simultaneously originated second mortgage identified by credit bureau data. This misrepresentation allows the borrower to take on additional debt, reducing their incentive to repay the loans and making the initial debt riskier. We identify second-lien misrepresentation by comparing loan-level mortgage data from BlackBox Logic with borrower-level credit report information from Equifax. Following the procedure in Piskorski et al. (2015), we first focus on first-lien loans with a merge confidence interval greater than or equal to 0.89. Second, we select loans with a new second-lien originating within one month before or after the first lien[5]. This filter, using

[5] Piskorski et al. (2015) uses a 45-day range, while Zhang et al. (2024) uses a one-month range. Since Equifax data identifies the close date of the second lien at the month level, we conservatively use a one-month

credit information, retains first-lien loans that truly have a simultaneous second lien. Third, to identify misreported second-lien loans, we require the loan to have a non-missing reported cumulative loan-to-value (CLTV) ratio within 1% of its loan-to-value (LTV) ratio. The small difference between the reported CLTV and LTV indicates that the borrower reports no simultaneous second lien.

2.2 Gambling Preference and Religiosity

Following Kumar et al. (2011), we construct county-level gambling preference based on religious data, specifically the ratio of Catholic residents to Protestant residents (CPRATIO). Although direct measures of local gambling preference are unavailable[6], we can infer the propensity by examining the proportion of different religious populations, which have distinct attitudes towards gambling according to their religious views. In the U.S., two widespread religions with differing views on gambling are Catholicism and Protestantism. While Protestant churches generally oppose gambling, Catholic churches maintain a tolerant attitude towards moderate levels of gambling. These differing views between the two religions are empirically supported to extend to financial markets (Kumar, 2009; Kumar et al., 2011; Han and Kumar, 2013; Chen et al., 2014). Therefore, regions with higher Catholic–Protestant ratios exhibit stronger gambling propensities.

We also construct county-level religiosity (REL) following Kumar et al. (2011). According to Hilary and Hui (2009), this measure of religiosity reflects risk attitudes, with higher values of REL indicating greater risk aversion. Moreover, religiosity captures social norms and ethical behavior, which play a significant role in deterring various types of mortgage misrepresentation (Conklin et al., 2022). Therefore, including the overall level of religiosity in the county ensures that our proxy for local gambling preference is independent of

range. A 45-day range can be achieved by assuming the close date is in the middle of the month. We also checked the 45-day range and found minor differences in the results.

[6]Gambling industry employment and establishment counts can serve as alternative measures of local gambling preference but have several limitations. We therefore use them only as robustness checks. See Appendix A for further discussion.

religion-induced risk aversion and ethical norms, thereby strengthening the potential causal interpretation of CPRATIO.

We use the dataset ”Longitudinal Religious Congregations and Membership File, 1980-2010 (County Level)” from ARDA to capture county-level geographical variation in religious composition. We sum the number of adherents in the religious traditional categories Evangelical Protestant, Mainline Protestant, and Black Protestant for the Protestant population and the number of adherents in the category Catholic for the Catholic population. We use the sum of adherents in all categories for the total religious adherents in the county. Since the data is available for each decade from 1980 to 2010, we follow previous studies to linearly interpolate the data to obtain values for missing years (Alesina and La Ferrara, 2000; Hilary and Hui, 2009; Kumar et al., 2011; Chen et al., 2014), and then calculate the CPRATIO (i.e., Catholic population to Protestant population) and REL (i.e., total religious adherents to total population) for each year.

2.3 Other Local Characteristics

In addition to gambling preference and religiosity, we also examine a rich set of local characteristics, covering both demographic factors and local economic conditions. Specifically, we use county-level data from 2000 on education, marriage, residential area, population, age, male-female ratio, and minority share, as well as zip code-level income information, from the U.S. Census Bureau. We further incorporate county-level crime statistics from 2005 to 2007 and voter registration data from 2006 obtained from the ICPSR. These demographic characteristics are closely related to risk preferences, ethical norms, cultural background, and financial literacy, all of which are likely to influence household financial decisions.

To capture local economic conditions that also play a crucial role in mortgage decisions, we employ county-level monthly unemployment rates from the U.S. Bureau of Labor Statistics. In addition, we use annual zip code-level house price indices (HPI) to construct measures of house price appreciation.

2.4 Mortgage Microdata

In addition to the second-lien misrepresentation measure, we also obtain other loan-level mortgage data from BlackBox Logic. The database includes a comprehensive set of loan characteristics at origination, such as the loan interest rate, borrower's FICO credit score, initial loan balance, loan-to-value ratio, amortization type (e.g., full amortization, interest-only, negative amortization), income documentation type (e.g., full, low-doc, or no-doc), interest rate type (e.g., fixed or adjustable), prepayment penalty, loan purpose (e.g., purchase or refinance), reported occupancy status (e.g., owner-occupied, investment, or second-home), delinquency method (i.e., MBA or OTS), and delinquency status (e.g., current, 30, 60, 90 days, etc.). The definitions of variables constructed from this information are presented in Table 1.

2.5 Descriptive Statistics

We study the effect of borrower characteristics on second-lien misrepresentation in a sample that includes loans without simultaneous second liens, loans with correctly reported simultaneous second liens, and loans with misreported simultaneous second liens. We select the sample period of 2005 to 2007, during which mortgage misrepresentation measures can be calculated using the databases and misrepresentations are not rare. Table 2 shows the summary statistics. For each continuous variable, we report the number of observations, mean, standard deviation, 25th percentile, 50th percentile, and 75th percentile. For dummy variables, we report the number of observations and mean. The continuous variables are winsorized at the 0.5 percent level to ensure our results are not driven by extreme values.

In our sample, we have 3,031,921 loans for which the borrower's area CPRATIO is available[7]. Among these loans, about 7.15 percent have misreported second liens, while 13.91 percent have reported second liens, resulting in a total of 21.06 percent. This percentage is

[7]We also exclude cases where the LTV is greater than 99% since our measure is calculated using LTV. An LTV greater than 99% is likely erroneous or highly unlikely to have a simultaneous second lien.

comparable to the one reported by Griffin and Maturana (2016) (10.2%), who used a different database and included all loans (honestly reported, fraudulently hidden, and truly without second liens). Figure 1 plots the nationwide distribution of the county-level proportion of second-lien misrepresentation in all loans, showing variations both across and within states.

Our primary independent variable of interest is local gambling preference (CPRATIO). While there are 3,090 available U.S. counties from 2005 to 2007, we only retain county-year level data where loan-level data is also available. For the 8,716 data points, the mean CPRATIO is 0.51 and the median is 0.19, indicating a large positive skewness[8].

Since we examine a rich set of local characteristics that may reflect borrowers' preferences, ethics, culture, and financial literacy, it is important to understand how these variables are correlated with one another. To assess this, we calculate pairwise correlations at the county-year level. Table 3 reports the results. CPRATIO and REL generally exhibit low correlations with other characteristics, suggesting that they capture distinct information not encompassed by other common local measures. Most other characteristics display low to moderate correlations. As expected, income and education are relatively highly correlated, since individuals with higher education levels typically earn higher incomes. Similarly, urban status and total population show a relatively strong correlation, consistent with the idea that counties in urban areas tend to be more densely populated. Nonetheless, because none of these correlations are high enough to raise concerns about multicollinearity, and since each variable captures a different aspect of borrower characteristics, we include all of them in the regressions.

[8]The mean and median CPRATIO from 1980 to 2005 are 0.60 and 0.23, respectively, in Kumar et al. (2011), which is quite close to our results if we include all counties.

3 Local Characteristics and Second-lien Misrepresentation

In this section, we first investigate the relationship between second-lien misrepresentation and local demographic characteristics that reflects household's feature in making financial decision. We conduct loan-level regression analysis of second-lien misrepresentation on the local characteristics using the following form of specification:

$$\begin{aligned}\text{Second-lien Misrepresentation} = \; & \alpha + \beta_1 \times \text{Local characteristics} \\ & + \beta_2 \times \text{Loan characteristics} + \text{Fixed effects}\end{aligned} \tag{1}$$

where *Second-lien Misrepresentation* is an indicator that equals one if the first-lien loan has simultaneous second-liens that are misreported at the loan application. *Local characteristics* includes the vector of all local characteristics, such as CPRATIO, REL, and education. *Loan characteristics* includes the vector of all loan-level information at the origination regarding the loan characteristics and borrower characteristics, such as interest rate, loan balance, and FICO score. *Fixed effects* include originator, state, and half-year fixed effects, each of which may influence mortgage misrepresentation[9]. We introduce these fixed effects sequentially into the regressions to assess whether the effects of local characteristics remain robust within and across groups. Because most local characteristic variables are measured at the county level, we cluster heteroskedasticity-robust standard errors by county in all specifications. All continuous variables are winsorized at the 0.5 percent level. Additionally, we standardized all the continuous variables so that the coefficient magnitude is comparable. Following Piskorski et al. (2015) and Yavas and Zhu (2024), we use OLS regressions in the main tests despite the binary nature of the dependent variable, and rely on probit models

[9]Although both originators and underwriters play important roles (Griffin and Maturana, 2016) and the related fixed effects are used in different literature, we control for originator fixed effects because originators are the agency that interact with the borrowers so that they may affect the decision of borrowers while underwriters do not have such influence in the borrowers decision-making process.

and causal forest approaches as robustness checks.[10]

Table 4 reports the regression results. Column (1) presents the baseline model with only local and loan characteristics. Column (2) adds originator fixed effects. Column (3) further includes state fixed effects, and Column (4) additionally incorporates half-year fixed effects.

We look at the results for each demographic characteristic that reflects the composition of the local population, focusing on their interpretation once other controls are included and testing their robustness in explaining second-lien misrepresentation. Among demographic characteristics, *CPRATIO*, which primarily serves as a proxy for gambling preference (Kumar et al., 2011), is consistently positive and significant, indicating that counties with stronger gambling tendencies experience higher misrepresentation. *REL*, which reflects religion-induced risk aversion and ethical norms (Hilary and Hui, 2009), is negative but fragile, suggesting that religious environments may discourage misrepresentation, though not robustly. *Education*, linked to financial literacy beyond income (Campbell, 2006), shows a positive but unstable association, consistent with better-educated households being more capable of exploiting financial opportunities. *Married*, reflecting household stability and homeownership propensity (Grinstein-Weiss et al., 2011), yields mixed and insignificant results, implying little systematic effect. *Income*, as an indicator of local financial strength, generally loads negatively once fixed effects are included, suggesting stronger financial conditions reduce incentives to misrepresent. *Urban*, capturing settlement patterns and loan access (Rogers, 2006), also turns negative with richer controls, pointing to greater misrepresentation in rural areas. *Over65*, the share of older residents who are typically risk-averse but borrowing constrained (Barsky et al., 1997; Nakajima and Telyukova, 2017), has a positive but weak effect, indicating older household may also participate in misrepresentation[11].

[10]The OLS model has an important advantage in allowing the inclusion of multiple fixed effects. By contrast, probit or logit models generally yield inconsistent estimates when more than one fixed effect is included in large samples (Wooldridge, 2010).

[11]This may contradict to common expectation on the sign due to high homeownership and greater risk averse. One potential explanation is that older households may have greater demand to lower their financial burden or extract more equity to support current spending. In our subsample analysis, the effect of *Over65* indeed only shows off for refinance and primary subsamples.

Male–female ratio, which reflects gender-induced risk attitudes (Croson and Gneezy, 2009), shows mixed and insignificant coefficients, implying that gender composition has little role in misrepresentation. *Minority*, linked to higher-cost mortgage terms and lower income levels (Bayer et al., 2018; Kennickell, 2009), also yields mixed and insignificant effects, suggesting racial composition does not systematically explain misrepresentation. *Part I crime*, reflecting serious violent and property offenses, tends to enter positively but without robustness, indicating that higher levels of social disorder may correlate with greater misrepresentation. *Part II crime*, associated with less severe but more pervasive offenses and thus more closely tied to everyday ethical norms, shows mixed and insignificant effects, suggesting that variation in local ethical climate has little systematic impact. Finally, *Democratic*, reflecting political culture and regulatory attitudes, generally loads negatively but not robustly, indicating that political leaning plays only a limited role in misrepresentation.

While not reflecting household composition, other local characteristics also affect the likelihood of misrepresentation. *Total population*, capturing county size, is significantly positive across all specifications, suggesting that more populous areas tend to exhibit higher rates of misrepresentation. *Unemployment*, measuring local labor market distress, is significantly positive in most cases but turns insignificantly negative once time fixed effects are included, implying that joblessness is generally associated with greater misreporting, though the effect varies across periods. *HPA*, reflecting past house price appreciation, shifts from significantly negative to significantly positive after the inclusion of time fixed effects, indicating that recent house price dynamics influence household leverage decisions, but the impact depends on comparisons within each time period.

Taken together, these results reveal a striking pattern: among all local demographics that could plausibly influence excessive leverage through mortgage misrepresentation, ranging from risk attitudes to financial literacy, only *CPRATIO* emerges as both robust and substantial in magnitude. A one standard deviation increase in *CPRATIO* raises the probability of second-lien misrepresentation by 0.14 percentage points, even after controlling for other local characteristics, detailed loan and borrower characteristics, and fixed effects.[12] This effect is also comparable to borrower-level fundamentals such as interest rate (-0.72 percentage points) and FICO score (-0.62 percentage points). By contrast, other demographics, such as education, religion, or income, are not robust across specifications or display much smaller coefficients. These findings motivate our interpretation of *CPRATIO* as a proxy for local gambling preference, which captures a distinct behavioral trait that systematically drives mortgage misrepresentation, whereas conventional demographics do not.

4 Interpreting the Effect of Gambling Preference: Borrower Behavior or Lender Practice

From the previous section, we find that second-lien misrepresentation is more likely to occur in counties with higher values of *CPRATIO*. We interpret *CPRATIO* as a measure of local gambling preference, following Kumar et al. (2011), who show that regions with higher *CPRATIO* exhibit stronger gambling and speculative tendencies. Gambling preference, in this context, refers to a willingness to accept a high probability of small losses for a small chance of a large gain (Markowitz, 1952). This behavioral tendency can influence financial decisions by making households more inclined to pursue opportunities with large potential payoffs despite substantial risks. In our setting, it implies that in areas with stronger gambling preferences, borrowers are more willing to take on excessive leverage through second-lien misrepresentation.

From the household perspective, misreporting a second lien creates an opportunity with asymmetric payoffs. The upside combines access to homeownership and the possibility of obtaining better loan terms. Homeownership carries both emotional and financial appeal. Emotionally, it is associated with the American Dream (Clinton, 1995; Bush, 2003), while

[12] The absolute effect appears small because the overall misrepresentation rate is only 7.15 percent, but it represents a 2 percent increase relative to the baseline.

financially, it offers the potential for wealth accumulation through house price appreciation and mortgage-interest tax deductions (Bostic and Lee, 2008; Goodman and Mayer, 2018). During the housing boom before the 2008 financial crisis, this upside appeared especially attractive because most households and lenders believed that house prices would continue to rise, which made the expected payoff distribution strongly right-skewed. Even if prices rose only moderately or stagnated, borrowers still expected to preserve some gains or at least maintain ownership. By contrast, the downside was relatively limited. The most common outcome of detection was loan denial, which entailed small application costs and minimal switching effort in an environment with abundant credit supply. Before the loan closed, lenders could file Suspicious Activity Reports, which sometimes led to investigation and charges. After closing, borrowers might face fines, restitution, or probation if misrepresentation was uncovered (Henning, 2009). In practice, however, law enforcement during this period focused on fraud for profit rather than fraud for housing (FBI, 2006, 2007). As a result, households who misreported to obtain housing rarely faced severe penalties. The asymmetry between large perceived gains and modest expected losses produces a right-skewed payoff structure that is particularly appealing to borrowers in areas with stronger gambling preferences.

However, the measure of gambling preference is constructed at the local level rather than the borrower level, which raises the possibility that the effect on second-lien misrepresentation may operate through lenders' practices rather than borrower decisions. Prior research has shown that both originators and underwriters can play significant roles in mortgage misreporting (Griffin and Maturana, 2016; Piskorski et al., 2015). These intermediaries were often aware of hidden second liens yet still allowed them to be misreported. More recently, by comparing portfolio loans with privately securitized mortgages, Yavas and Zhu (2024) provide evidence that second-lien misrepresentation tends to arise in the early stages of intermediation by lenders rather than at the underwriting stage.

Therefore, it is necessary to assess whether the observed relationship should be attributed primarily to borrower behavior or to lender practices. In the remainder of this section, we examine evidence from both perspectives and show that the patterns are more consistent with an interpretation based on borrower behavior.

4.1 Evidence from Borrower Behavior

According to Christensen et al. (2018), the impact of gambling preference on misreporting should be more pronounced in situations where individuals face stronger incentives. Since our interpretation primarily emphasizes the borrower's perspective, we examine whether the effect is amplified when borrowers' incentives are greater, using subsample analysis.

We begin by examining subsamples defined by loan-level characteristics that directly capture borrower incentives. Specifically, we partition the sample by the loan's property occupancy type (primary or non-primary) and by the loan's purpose type (purchase or refinance). A primary occupancy type indicates that the property is owner-occupied, which is tied to the borrower's need for stable housing and the utility derived from living in the home. In contrast, a non-primary occupancy type implies that the property is used as a second home or investment, suggesting that the borrower already owns a residence and is motivated by discretionary consumption or wealth accumulation. Because the incentive to misrepresent should be greater when stable housing is at stake, we expect the effect of gambling preference is more evident in the primary occupancy subsample. Similarly, if the loan purpose is a purchase, the borrower does not yet own the property and must obtain it to begin consumption or investment. By contrast, if the purpose is refinance, the borrower already owns the property and seeks to improve financial conditions by lowering costs or accessing home equity. In this case, the incentive to misrepresent should be stronger in purchase loans, making them a setting where gambling preference is expected to play a larger role.

Table 5 reports the subsample results. Columns (1) and (2) present estimates for the primary and non-primary groups. Most observations in the overall sample fall into the primary category. Moreover, only the coefficient of CPRATIO in the primary subsample is significant, and its economic magnitude is substantially larger than that in the non-primary group. This finding suggests that borrowers with stronger gambling preferences are more likely to misrepresent second liens when the property is intended as a primary residence, consistent with our expectation that the relationship between gambling preference and misrepresentation is stronger in the primary subsample, where borrower incentives are greater.

Columns (3) and (4) present the results for the purchase and refinance subsamples. About 43 percent of the observations come from the purchase group. The coefficient of CPRATIO in the purchase subsample is significant at the 1 percent level, whereas it is only marginally significant in the refinance subsample. Additionally, the economic magnitude in the purchase group is much larger than in the refinance group. This indicates that borrowers with stronger gambling preferences tend to misrepresent second liens primarily when purchasing a house, reinforcing our expectation that the relationship between gambling preference and misrepresentation is stronger in the purchase subsample, where borrower incentives are greater.

We also conduct subsample analyses based on characteristics that not only relate to broader debates about the origins of the financial crisis but also capture variation in borrowers' incentives. Specifically, we partition the sample by area income (high or low income), credit score (high or low FICO), and past house price appreciation (high or low HPA). The high- and low-income subsamples are defined by comparing local household median income with the sample median at the ZIP code level. The high- and low-credit-score subsamples are divided using a FICO cutoff of 670[13]. The high- and low-HPA subsamples are defined based on the state-year median HPA, calculated from all available ZIP code-level data.

Area income and credit score are closely tied to the credit expansion argument, which centers on whether the expansion of mortgage credit leading up to the crisis was widespread across the population. In our setting, gambling preference should affect all groups, as all

[13]670 is the boundary between fair and good levels as evaluated by the institution. We also tried the median of the sample, 682, which yielded similar results.

contributed to the crisis. However, low-income and low-credit-score borrowers are expected to show stronger effects because they have greater incentives to misrepresent, stemming from tighter financial constraints (low income) or more limited credit access (low FICO). As shown in Table 6, columns (1) to (4), the economic magnitude of CPRATIO is large across different groups[14], and the effects are larger and more significant in the low-income and low-FICO subsamples. This pattern is consistent with the credit expansion view and also reinforces the interpretation of borrower gambling preference.

Past house price appreciation is closely related to the belief distortion channel, which emphasizes that households during this period held overly optimistic expectations of future price growth. In our context, households in areas with higher past appreciation are more likely to extrapolate those increases into the future (Shiller, 2007), creating stronger incentives to misrepresent due to higher expected returns. As shown in Table 6, columns (5) and (6), the coefficient of CPRATIO is larger in magnitude and more statistically significant in high-HPA areas. This evidence indicates that our findings are also consistent with the belief distortion channel, while continuing to support the interpretation of borrower gambling preference.

Overall, the subsample analysis shows that second-lien misrepresentation rises more sharply with gambling preference when borrowers face stronger incentives. This evidence support our interpretation of the effect from borrower's perspective.

4.2 Evidence from Lender Practice

To assess this issue from the perspective of lenders' practices, we employ the setting of ease of securitization around a FICO score of 620, as described by Keys et al. (2010). Due to the underwriting guidelines established by government-sponsored enterprises, Fannie Mae and Freddie Mac, low documentation loans made to borrowers with FICO scores of 620 or higher are much easier to securitize. Consequently, lenders have lax screening standards

[14] Recall that in the baseline results, the coefficient of CPRATIO is about 0.0014.

for such loans, and a regression discontinuity design could reveal an upward jump in the default rate for low documentation loans. Additionally, Griffin and Maturana (2016) find that the second-lien misrepresentation rate also jumps at this cutoff, implying that this type of misrepresentation is associated with lenders' incentives to securitize the loan.

Utilizing this shock to lenders, we first study the extent to which lenders facilitate second-lien misrepresentation by comparing loans with misreported simultaneous seconds to other types of loans. Specifically, we examine the jumps in the number of loans and the jumps in the default rate for all loans, loans with correctly reported simultaneous seconds, and loans with misreported simultaneous seconds. If lenders specifically facilitate second-lien misrepresentation, then the jump ratio in the number of loans with misreported simultaneous seconds should be greater than for other types of loans. Additionally, if such special facilitation of misrepresentation stems from lenders' screening efforts, the jumps in the default rate for loans with misreported simultaneous seconds should be significantly greater than those for loans with correctly reported simultaneous seconds. In contrast, if the jump ratio in the number of loans with misreported simultaneous seconds is similar to or smaller than for other types of loans, lenders do not specifically facilitate second-lien misrepresentation but increase it with the increase of other loans. Finally, if the jumps in the default rate for loans with misreported simultaneous seconds are no larger than those for loans with correctly reported seconds, this would imply that lenders' screening effort does not play a distinct role in facilitating misrepresentation.

Building on this general picture, we next investigate whether the effect of gambling preference on second-lien misrepresentation aligns with lender practices. Since the variation in second-lien misrepresentation related to gambling preference could come from either lenders or borrowers, and the shock we employed is only for lenders, if the increase in second-lien misrepresentation in high gambling preference areas is smaller, it indicates that lenders play a smaller role in increasing misrepresentation in high gambling preference areas. This implies that the positive relationship between gambling preference and second-lien misrepresentation is mainly attributed to borrowers. Specifically, we examine the jump ratio in the number of loans and the jumps in the default rate to understand how lender's roles differ across areas. If the effect of local gambling preference on second-lien misrepresentation is in line with lenders' practice (i.e., lenders make second-lien misrepresentation more prevalent in high gambling preference areas), this shock to lenders would lead to different outcomes between high and low gambling preference areas, favoring high gambling preference areas (i.e., a greater jump ratio in the number of loans with misrepresentation and a larger increase in the default rate). In contrast, a smaller jump ratio in the number of misreported loans and the default rates of misreported loans in high gambling preference areas indicates that lenders did not facilitate more misrepresentation in such areas.

We take RDD approach using local linear regressions. The credit scores are normalized as follows:

$$\text{C} = \text{Credit Score} - \text{Threshold} \tag{2}$$

where $Threshold$ is 620 for low documentation loans. Then, we define D to distinguish credit scores that are over or below the threshold as follows:

$$\text{D} = \begin{cases} 1, & if\ \text{C} \geq 0 \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

To differentiate the effect between high and low local gambling preference areas, we define high local gambling preference ares:

$$\text{G} = \begin{cases} 1, & if\ \text{CPRATIO} \geq 1.2921 \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

We use 1.2921 as the cutoff for two reasons[15]. First, only 10 percent of county-years in our sample have a CPRATIO greater than or equal to 1.2921, indicating that areas with

[15] We also tried other reasonable cutoffs, such as 1, and the results are robust.

such a CPRATIO indeed have a high local gambling preference. Second, about half of all loans in our sample are originated in these areas, making our findings about high and low gambling preference areas comparable.

To apply the RDD approach using local linear regressions, we choose a bandwidth of 10, which corresponds to a FICO score range of 610 to 630. We selected this window because it ensures that no other jumps are found in the literature[16] and it is close to the optimal bandwidth generated by a data-driven bandwidth selection method[17] (Calonico et al., 2020). We use the fixed bandwidth rather than the optimal bandwidth so that the RDD results in different cases are comparable[18]. For the default rate and number of loans in high and low gambling preference areas, we estimate the four cases separately using the following specification:

$$Y = \alpha + \beta D + \gamma C + \delta D \times C + \epsilon \tag{5}$$

where Y is the default dummy variable for loan i originated at time t or the number of loans at each FICO score. In our baseline model, we use uniform kernel for estimation.

Finally, to calculate the difference in the discontinuities (diff-in-disc) between high and low gambling preference areas, we follow Dickert-Conlin and Elder (2010) and Grembi et al. (2016) to estimate the following model:

$$Y = \beta_0 + \beta_1 D + \beta_2 C + \beta_3 D \times C + \beta_4 G + \beta_5 G \times C + \beta_6 G \times D + \beta_7 G \times C \times D + \epsilon \tag{6}$$

where β_6 is the parameter that measures the difference in the discontinuity in Y at credit score cutoff for high versus low gambling preference areas. When we apply this equation to calculate the difference in jump ratios between high and low gambling preference areas,

[16] For example, FICO scores of 600 and 660 could be other potential cutoffs for jumps.
[17] The optimal bandwidth for the number of all loans is 10.50.
[18] Using the optimal bandwidth does not affect our findings.

we first rescale the data to obtain the t-statistics. We divide the data in high gambling preference areas by the estimated value at FICO 620^{-} in those areas. Similarly, we divide the data in low gambling preference areas by their corresponding estimated value at FICO 620^{-}. After this rescaling, the discontinuity estimated in high or low gambling preference areas can be directly interpreted as a multiple of its corresponding estimated value at FICO 620^{-}. The difference between the two discontinuities represents the difference in jump ratios between high and low gambling preference areas.

We first look at the jumps in the number of loans and the jumps in the default rate in all areas, examining the pattern of second-lien misrepresentation given a shock to lenders. Panel A of Table 7 presents the results for jumps in the total number of loans, loans with correctly reported simultaneous seconds, and loans with misreported simultaneous seconds. Overall, the total number of low-documentation loans doubles from 620^{-} to 620^{+}, consistent with the findings in Keys et al. (2010). Loans with simultaneous seconds increase more than those without, whether correctly reported or misreported. This shows that the ease of securitization is indeed applicable to such loans. Moreover, as shown in Panel A of Figure 2, the jump ratio of misreported simultaneous seconds is less than the jump ratio of correctly reported simultaneous seconds, indicating that lenders do not facilitate second-lien misrepresentation more than they do correctly reported seconds.

Panel B of Figure 2 compares the jump in default rates between correctly reported and misreported seconds. The figure shows that the increase in default rates is less pronounced for loans with misreported seconds than for those with correctly reported seconds. Panel B of Table 7 reports regression estimates with control variables.[19] Consistent with previous literature, there is a jump of 0.039 in the default rate for all loans, indicating overall reduced screening for 620^{+} loans. This lax screening is also observed in loans with correctly reported simultaneous seconds, with a statistically significant jump of 0.045. For the loans with mis-

[19]Figures 2 and 3 present unconditional results with conventional t-statistics to illustrate the pattern of jumps in default rates relative to their levels. Tables 7 and 8 present conditional results with bias-corrected t-statistics for more precise estimation.

reported simultaneous seconds, the jump in the default rate is similar in magnitude but statistically insignificant, suggesting that the lax screening effect is not as pronounced as for correctly reported seconds. These findings illustrate that lenders facilitate loans with simultaneous second liens, whether correctly reported or misreported, and that the lax screening effect is no larger for misreported seconds than for correctly reported seconds.

After looking at the jumps in all areas, we further explore the patterns between high and low gambling preference areas to determine if the effect aligns with lender practices. Table 8 shows the results for the number of loans in Panel A and the results for default rates in Panel B. For the number of loans, the magnitude of increases is similar between high and low gambling preference areas for all loans and loans with correctly reported simultaneous seconds. This indicates no systematic difference between high and low gambling preference areas in ease of securitization. In contrast, as shown in Panel A of Figure 3, the magnitude of the increase in the number of loans with misreported seconds in high gambling preference areas is much smaller than in low gambling preference areas. Given that the ease of securitization is a shock to lenders (Keys et al., 2010) and lenders facilitate misrepresentation with the objective of securitization (Griffin and Maturana, 2016), this result suggests that lenders do not play a larger role in increasing second-lien misrepresentation in high gambling preference areas[20].

For default rates, the jumps between high and low gambling preference in all types of loans do not show significant differences. Additionally, for loans with misreported simultaneous seconds, the jump in high gambling preference areas is smaller than in low gambling preference areas, as shown in Panel B of Figure 3, although the difference is not significant. This suggests that lenders in high gambling preference areas are not more likely to facilitate misrepresentation by having laxer screening in such areas. Therefore, taken together with

[20]Since borrowers play a larger role in increasing second-lien misrepresentation in high gambling preference areas, lenders' effect in increasing the number is more limited, making the magnitude of the jump smaller. In contrast, since borrowers play a smaller role in increasing second-lien misrepresentation in low gambling preference areas, lenders' effect in increasing the number is less limited, making the magnitude of the jump bigger.

the evidence on the number of loans, these findings suggest that lenders in high gambling preference areas are not more inclined to increase such misrepresentation than lenders in low gambling preference areas. In other words, the evidence does not support the interpretation that the effect operates through lenders' practices.

5 Outcomes of Gambling Preference-Related Misrepresentation

In the previous section, we found a significant correlation between gambling preference and mortgage misrepresentation. Our next step is to investigate the outcomes of gambling preference-related mortgage misrepresentation. Prior studies show that mortgage misrepresentation generally leads to worse loan performance (Piskorski et al., 2015; Griffin and Maturana, 2016), but it is unclear whether the misrepresentation associated with gambling preference has systematically different consequences. Since gambling is typically linked to adverse financial outcomes, we conjecture that gambling preference-related misrepresentation exacerbates loan performance problems. We conduct loan-level regression analysis of delinquency on second-lien misrepresentation and gambling preference using the following form of specification:

$$\begin{aligned}\text{Default} = \; & \alpha + \beta_1 \times \text{Second-lien Misrepresentation} + \beta_2 \times \text{Gambling Preference} \\ & + \beta_3 \times \text{Second-lien Misrepresentation} \times \text{Gambling Preference} \\ & + \beta_4 \times \text{Controls} + \text{Fixed effects}\end{aligned} \tag{7}$$

where *Default* is an indicator that equals one if the loan becomes 90 days or more delinquent using MBA method. *Second-lien Misrepresentation* is an indicator that equals one if the first-lien loan has simultaneous second-liens that are misreported at the loan application. *Gambling Preference* is local gambling preference measured by $CPRATIO$. *Second-lien*

Misrepresentation × *Gambling Preference* is the interaction term between second-lien misrepresentation and gambling preference. *Controls* includes the vector of all other variables used in the baseline findings, such as local characteristics and loan-level information. *Fixed effects* include originator, state, and half-year fixed effects. We include control variables, fixed effects, and standard error clustering in all specifications, but we gradually include the mortgage misrepresentation term, gambling preference term, and the interaction term.

The results are reported in Table 9. Column (1) includes the misrepresentation indicator only and confirms that loans with misreported second liens default at significantly higher rates than other loans. Column (2) adds the gambling preference measure as a baseline control. Column (3) introduces the interaction term between misrepresentation and gambling preference, which is the main focus of our analysis.

The coefficient on second-lien misrepresentation is significantly positive across all specifications, indicating that loans with misreported seconds default at higher rates. The interaction between misrepresentation and gambling preference is also positive and significant, showing that the adverse default effect of misrepresentation is stronger in counties with higher gambling preference. This suggests that gambling preference amplifies the negative performance consequences of misrepresentation.

To further illustrate the economic magnitude of our estimates, we do a back-of-the-envelope calculation. A one-standard-deviation increase in CPRATIO raises the default probability of misreported loans by about 1.4 percentage points. Given 216,736 such loans nationally in the sample, this translates into roughly 3,034 additional defaults. At an average first-lien balance of approximately $262,000, this corresponds to about $0.8 billion in first-lien balance for private-label RMBS during the crisis period. This magnitude illustrates that the interaction of gambling preference and mortgage misrepresentation is not only statistically significant but also economically meaningful.

6 Robustness

6.1 Nonlinear Model

The results in the baseline model are estimated using a linear probability model (OLS). To ensure our findings are not driven by this modeling choice, we also use a nonlinear specification (probit) for inference. We present the marginal effects in Table 10. Since we control for simultaneous second liens in the regression, the probit model drops all observations without simultaneous seconds due to perfect failure prediction of misrepresentation. The marginal effects of CPRATIO remain significant while other demographics related to household risk preference, ethics, culture, and financial literacy continue to show little robustness.

A potential concern with this approach is that the sample is limited to loans with simultaneous seconds, so the estimated misrepresentation rate no longer reflects the entire loan population. To address this issue, we re-estimate the probit model without the loan-level indicator for simultaneous seconds. Instead, we control for the prevalence of simultaneous seconds at the county level. The marginal effects from this specification are reported in Table 11. Once again, *CPRATIO* remains robustly significant, while the significance of other demographics changes somewhat depending on the modeling choice.

6.2 Causal Forest and Causality

To enhance the causal inference between gambling preference and mortgage misrepresentation and to explore the heterogeneous treatment effects of mortgage misrepresentation on mortgage default conditioned on gambling preference, we employ the causal forest approach proposed by Wager and Athey (2018). This method uses an augmented inverse propensity-weighted estimator with the random forest method in machine learning, incorporating an honesty condition. It provides double robustness (compared to propensity score matching) and high efficiency for high-dimensional models (compared to nearest-neighbor matching) in observational settings[21]. This approach has also been used in finance, such as in Rampini and Viswanathan (2022) for secured debt and Gulen et al. (2021) for corporate finance, and has shown better performance than traditional causal inference designs (Gulen et al., 2021) in terms of accuracy.

We first use this approach to investigate the causality between gambling preference and mortgage misrepresentation. In the causal forest approach, we set mortgage misrepresentation as the outcome and CPRATIO as the treatment. We use all control variables from Table 4 as matching variables to help grow trees and forests[22]. All continuous variables, including CPRATIO, are winsorized at the 0.5 percent level and then standardized. Since fixed effects are not applicable in this approach, we create a variable for half-year periods, starting from the first half of 2005 as 1, to account for potential time effects. Similar to Athey and Wager (2019), who cluster observations by school ID, we cluster observations by state but give each unit the same weight so that larger clusters receive more weight[23]. By using such cross-fitting, our results do not solely come from any single state, but states with a greater number of observations do have greater weights. To balance computational capacity and the accuracy of confidence intervals, we grow 2000 trees for the forest, which is also the default setting in Athey and Wager (2019). For the honesty property, we use the default splitting fraction: for each sample, we use 50 percent of the data for splitting and the remaining data for estimation. Table 12 shows the results using the causal forest. The coefficients in the regressions of second-lien misrepresentation remain significant, and the magnitude is similar to those estimated by OLS.

Second, we study the heterogeneous treatment effects of mortgage misrepresentation on default conditioned on different levels of gambling preference. We set default as the

[21] See Wager and Athey (2018) for statistical illustration. See Athey and Wager (2019) for an example application.

[22] We require observations to be non-missing for all variables.

[23] Due to the limitation of the grf package in R, which requires the matching variables to be numerical, we do not create a numerical variable for originators to avoid falsely treating closer values as shorter distances. We also do not cluster observations by originator because the large number of originators would excessively increase the computational burden.

outcome and mortgage misrepresentation as the treatment. Except for adding CPRATIO to the matching variable matrix, the other matching variables and clustering variables are the same as those in the above causal forest regressions. For the same considerations, we also grow 2000 trees for the forest and use 50 percent of the data in each sample for splits. After estimating the causal forest, the average treatment effects of mortgage misrepresentation on default are calculated for each quarter of the sample sorted by CPRATIO. Figure 4 shows the results for second-lien misrepresentation. The treatment effects are generally larger when CPRATIO is greater, which is consistent with the results from the OLS model.

6.3 Alternative Gambling Preference Measures

While CPRATIO serves as our primary measure of gambling preference, we also employ alternative measures to address concerns that CPRATIO may reflect broader religious differences rather than gambling-specific tendencies. Following studies such as Markham et al. (2023), which use the density of gambling outlets as a measure of gambling exposure, we incorporate employment of gambling establishments as alternative measures that reflect gambling preference through local supply. Specifically, we utilize per capita employment in the gambling industry at the core-based statistical area (CBSA) level. This alternative helps isolate local gambling behavior more directly. We regress the second-lien misrepresentation dummy variable on the alternative measure, alongside other local controls at the CBSA level, loan-level controls, and fixed effects (i.e., state, lender, and half-year). Additionally, we include controls for employment in casino hotels to account for tourist-driven gambling demand[24]. We also include nonemployer gambling establishments to capture complementary gambling demand that may be excluded by the primary measures due to database limitations. See Appendix for detailed descriptions of the data and empirical tests.

Table 13 presents the results of the primary regressions. Columns (1) report specifi-

[24]In the CBP database, casino hotels are categorized separately from the gambling industry. Because casino hotel employees also provide hotel-related services beyond gambling, it is necessary to control for casino hotels separately.

cations using per capita employment in the gambling industry. Columns (2) include both employment metrics for the gambling industry and casino hotels, as well as nonemployer gambling establishments. The regression results indicate that employment metrics for the gambling industry are significantly positive, suggesting a strong link between local gambling preference and the likelihood of second-lien misrepresentation. By contrast, casino hotel variables, which reflect a mix of local and tourist demand, yield coefficients that are generally insignificant. Although nonemployer gambling businesses primarily reflect local demand, they tend to offer limited gambling access, and their associated coefficients are likewise insignificant. Overall, we find that per capita employment measure in the gambling industry, which is used as alternative proxy for local gambling preference, supports the hypothesis that second-lien misrepresentation is more prevalent in regions with elevated gambling preference.

6.4 RDD and Diff-in-disc concerns

Since the purpose of RDD is to show that lenders do not specifically facilitate second-lien misrepresentation compared to loans with simultaneous seconds in general, we need to verify that the underlying borrower and collateral fundamentals are smooth at the cutoff. Continuity in these pre-determined variables ensures that borrowers above and below the cutoff are drawn from comparable populations, so that differences in the probability of having a second lien or misreporting one can be attributed to changes in securitization incentives rather than to shifts in borrower quality[25]. Table 14 presents the results, and all local characteristics, such as CPRATIO and REL, and collateral fundamentals, such as LTV and balance, are smooth around the cutoff.

[25]This approach differs from Keys et al. (2010), who use the 620 cutoff to establish a causal link between ease of securitization and lenders' screening effort, and from Yavas and Zhu (2024), who argue that lenders' misreporting itself was a direct consequence of securitization. Our purpose is instead to benchmark misreporting against the facilitation of second-lien origination, showing that both margins respond similarly to securitization incentives. In this setting, the relevant validity check is continuity in borrower and collateral fundamentals, while discontinuities in contract terms, such as interest rates, ARM versus FRM incidence, prepayment penalties, negative amortization features, cash-out refinancing, or occupancy type, reflect lender responses to securitization demand and should be interpreted as part of the treatment channel rather than as threats to the RDD design.

Our main evidence regarding the interpretation from lenders' practices is from the the comparison of high and low gambling preference. To validate our application of the diff-in-disc approach in comparing loan volumes and default rates between high and low gambling preference areas, we address several methodological concerns. First, we examine and test the three core assumptions outlined by Grembi et al. (2016) regarding the implementation of the diff-in-disc framework. Second, we assess the sensitivity of our results to key modeling choices, including the inclusion of control variables, bandwidth selection, and estimation kernel specification. Our findings remain robust across these variations. Third, we perform a placebo test to confirm that the observed effects are not driven by random variation. For full details of these robustness checks and supporting evidence, see Appendix. Overall, these tests confirm the validation of the diff-in-disc approach and provide evidence that does not support the interpretation of the gambling preference effect through lenders' practices.

6.5 Misrepresentation Rate

When exploring the effect of gambling preference from the perspective of lenders' practices, we use the jumps in the number of loans. The advantage of using this measure compared to the misrepresentation rate is that it allows us to study changes in different types of loans without the complication of shifts in both the numerator and denominator. However, since the number of loans is aggregated, one concern is that this approach does not employ control variables as in the baseline regression, making the results subject to differences in controls. To address this issue, we also examine the jumps in the rate of having correctly reported seconds and the rate of having misreported seconds, estimated with the same control variables as in the baseline regressions.[26] Table 15 presents the results for the raw rate, the rate after controlling for all other variables, and the rate after controlling for all other variables and fixed effects. Reported seconds are similar between high and low gambling preference areas, with the high areas showing a slightly greater jump. In contrast, misreported seconds are

[26]We do not calculate the rate at the county level. Instead, we use the dummy of second-lien misrepresentation directly so that we can include loan-level controls as in the baseline model.

smaller in high areas, although the difference may not be statistically significant once all controls and fixed effects are included. These results provide evidence that does not support the interpretation based on lenders' practices, under which the jump in misrepresentation should be greater in high gambling preference areas.

6.6 Subsample Evidence on Outcomes of Gambling Preference

While Section 5 shows that gambling in mortgage applications generally leads to worse outcomes, we now examine whether this effect varies across subsamples.

Table 16 reports results for subsamples related to the credit expansion and belief distortion arguments. The interaction terms, which measure how default rates change with gambling preference conditional on the occurrence of second-lien misrepresentation, are economically meaningful and statistically significant across all subsamples. These findings indicate that regardless of income class or the extent of belief distortion, when misrepresentation is associated with high gambling preference, loan outcomes are worse.

6.7 Alternative Measures of Default

Borrowers can default on their mortgages for various reasons over different time horizons. Our measure of default uses the MBA method of delinquency for 90 days or more and is restricted to the first three years after origination. By adjusting the definition to be more restricted (fewer default cases) or less restricted (more default cases), we can investigate whether the effect found in section 5 is generally applicable. Therefore, we consider the following default measures: 90 days or more delinquency using the MBA method in the first two years after origination (more restricted), 60 days or more delinquency using the MBA method in the first three years after origination (less restricted), and bankruptcy/foreclosure/REO in the first three years after origination (more restricted).

Table 17 presents the results of different default measures. We also include the results of the original measure in column (1). The significance of second-lien misrepresentation and its interaction with CPRATIO are robust. Additionally, when the default measure is less restricted, the effect of mortgage misrepresentation is greater. The effect of gambling preference-related mortgage misrepresentation is also greater when the default measure is less restricted.

7 Conclusion

High household leverage was an important feature of the housing boom preceding the financial crisis. A particularly striking form of excessive leverage taking was the misrepresentation of second liens, which allowed borrowers to overstate their borrowing capacity and obtain mortgages beyond sustainable limits. This paper examines how borrower heterogeneity shaped the decision to take on such high leverage.

Using a religion-based proxy for gambling attitudes, we find that second-lien misrepresentation is more prevalent in gambling-prone counties. The relationship is especially strong when borrower incentives to misreport are high. In contrast, evidence from the FICO 620 securitization cutoff provides little support for lender-based explanations. We also show that the negative performance consequences of misrepresentation are amplified in high-gambling-preference areas. Taken together, the results suggest that borrower behavioral traits, specifically gambling preference, played a significant role in fueling excessive leverage before the crisis. Recognizing the influence of such traits is important for understanding the fragility of mortgage markets and the behavioral foundations of financial instability.

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Table 1
Variable Definitions

Variable name	Definition
Mortgage Reporting	
Misreported second	Indicator that equals one if the borrower has simultaneous second-liens and misreported the second-liens on the loan application
Reported second	Indicator that equals one if the borrower has simultaneous second-liens and correctly reported the second-liens on the loan application
Simul. second	Indicator that equals one if the borrower have simultaneous second-liens regardless the reporting status on the loan application
Local Characteristics	
CPRATIO	Ratio of the county's Catholic residents to Protestant residents
REL	Proportion of the county population that are religious adherents
Education	Proportion of the county population over age 25 that has completed a bachelor's degree or higher
Married	Proportion of the county population over age 15 that is married
Income	Natural logarithm of the zip code level median household income
Urban	Proportion of the county population that lives in urban area
Total population	Natural logarithm of the county total population
Over65	Proportion of the county population over age 65
Male-female ratio	Ratio of the county's male residents to female residents
Minority	Proportion of the county non-white residents
Part I crime	Crime rates of part I offenses expressed as the number of crimes per 10,000 residents per year
Part II crime	Crime rates of part II offenses expressed as the number of crimes per 10,000 residents per year
Democratic	Indicator that equals one if the Democratic partisanship index is equal or greater than 50%
HPA	Zip code house price appreciation in the two years prior to loan origination year. County level data used if zip code index is not available
Unemployment	Unemployment rate of the county
Loan characteristics	
Interest rate	Loan interest rate at origination
FICO	Natural logarithm of borrower's FICO credit score at loan origination
Balance	Natural logarithm of the initial loan balance
LTV	Loan-to-value ratio at loan origination
ARM	Indicator that equals one if the loan is an adjustable rate mortgage
Option ARM	Indicator that equals one if the loan has an ARM convertibility clause
Negative amortization	Indicator that equals one if the loan allows negative amortization
Low or no doc.	Indicator that equals one if the loan is originated with no or limited documentation
Prepayment penalty	Indicator that equals one if the loan would be assessed a penalty on any early voluntary prepayment
Cash-out	Indicator that equals one if the loan purpose is cash out refinancing
No-cash-out	Indicator that equals one if the loan purpose is no cash out refinancing
Investment	Indicator that equals one if the loan occupancy status is investment
Second-home	Indicator that equals one if the loan occupancy status is second-home
Default	Indicator that equals one if the loan becomes 90 days or more delinquent using MBA method in the first three years after origination

The table reports the variable definitions used in the empirical analysis part.

Table 2
Descriptive Statistics

	N	Mean	SD	P10	P25	P50	P75	P90
Simultaneous second liens (loan level)								
Misreported second (%)	3,031,921	7.15	25.76					
Reported second (%)	3,031,921	13.91	34.61					
Simul. second (%)	3,031,921	21.06	40.77					
Loan characteristis (loan level)								
Interest rate (%)	3,026,958	6.41	2.12	2.75	5.88	6.50	7.55	8.75
FICO	2,993,630	681.78	72.60	578.00	632.00	688.00	739.00	776.00
Balance (ln)	3,031,921	12.40	0.73	11.44	11.87	12.40	12.98	13.31
LTV (%)	3,031,921	74.82	13.34	56.65	70.00	80.00	80.00	90.00
ARM (%)	3,031,921	58.87	49.21					
Option ARM (%)	3,031,921	0.24	4.84					
Negative amortization (%)	3,031,921	13.58	34.26					
Low or no doc. (%)	3,031,921	65.42	47.56					
Prepayment penalty (%)	3,031,921	40.99	49.18					
Cash-out (%)	3,031,921	40.66	49.12					
No-cash-out (%)	3,031,921	14.73	35.44					
Investment (%)	3,031,921	10.27	30.36					
Second-home (%)	3,031,921	3.55	18.49					
Default (%)	3,014,278	23.51	42.40					
Local characteristics (county-year level)								
CPRATIO	8,716	0.51	0.88	0.01	0.04	0.19	0.54	1.29
REL (%)	8,716	59.59	17.64	36.62	45.90	58.51	72.42	89.21
Part I crime (per 10,000 residents)	8,228	281.08	151.18	113.86	162.60	246.71	360.25	501.30
Part II crime (per 10,000 residents)	8,710	363.32	214.02	97.02	222.21	348.11	485.28	628.43
Local characteristics (county-month level)								
Unemployment (%)	71,156	5.14	1.73	3.20	3.90	4.90	6.00	7.50
Local characteristics (county level)								
Education (%)	3,035	16.64	7.51	9.70	11.20	14.40	19.30	26.60
Married (%)	3,035	60.39	5.10	53.90	57.90	61.30	63.90	66.00
Urban (%)	3,035	40.62	30.54	0.00	12.93	40.32	64.65	84.67
Total population (ln)	3,035	10.49	1.11	9.52	9.52	10.15	11.05	12.09
Over65 (%)	3,035	14.71	4.05	9.90	12.10	14.40	17.00	20.00
Male-female ratio	3,035	0.98	0.06	0.92	0.94	0.97	1.00	1.05
Minority (%)	3,035	15.32	15.80	2.02	3.34	8.71	22.96	37.90
Democratic	3,015	22.79	41.95					
Geographic characteristics (zipcode-year level)								
HPA	76,535	18.45	14.28	4.23	7.55	13.93	26.56	39.99
Geographic characteristics (zipcode level)								
Income (ln)	26,517	10.57	0.34	10.17	10.34	10.53	10.77	11.02

The table presents descriptive statistics for the variables used in our study, covering the sample period from 2005 to 2007. For all continuous variables, we report the number of observations, mean, standard deviation, 25th percentile, 50th percentile, and 75th percentile. For all dummy variables, we report the number of observations, mean, and standard deviation. The variables are winsorized at the 0.5 percent level. (ln) indicates that the value of the variable is the natural logarithm of the original value. (%) indicates that the value of the variable is expressed as a percentage.

Table 3
Local Characteristics Variables Correlation

	CPRATIO	REL	Income	Education	Married	Urban	Total pop.	Over65	Male-female	Minority	Part I	Part II	Democratic	UE	HPA
CPRATIO	1.00														
REL	-0.01	1.00													
Income	0.26	-0.11	1.00												
Education	0.26	-0.10	0.64	1.00											
Married	-0.19	0.12	0.13	-0.30	1.00										
Urban	0.32	-0.01	0.41	0.52	-0.40	1.00									
Total population	0.32	-0.18	0.50	0.53	-0.37	0.74	1.00								
Over65	-0.15	0.31	-0.42	-0.34	0.23	-0.33	-0.37	1.00							
Male-female ratio	0.05	-0.21	0.12	-0.01	0.18	-0.17	-0.18	-0.21	1.00						
Minority	0.08	-0.09	-0.15	0.02	-0.52	0.20	0.23	-0.30	-0.12	1.00					
Part II crime	0.01	-0.13	-0.01	0.14	-0.44	0.49	0.47	-0.26	-0.19	0.43	1.00				
Part I crime	0.01	0.03	-0.08	0.00	-0.21	0.19	0.07	-0.20	-0.09	0.18	0.33	1.00			
Democratic	0.28	-0.07	0.05	0.18	-0.43	0.18	0.25	-0.07	-0.09	0.29	0.19	0.07	1.00		
Unemployment	-0.06	-0.20	-0.34	-0.40	-0.14	-0.17	-0.08	0.02	-0.07	0.22	0.13	0.13	0.17	1.00	
HPA	0.26	-0.29	0.18	0.23	-0.08	0.15	0.23	-0.01	0.11	0.14	0.02	-0.06	0.09	-0.21	1.00

The table shows the correlation between local characteristics variables at the county-year level. Income and HPA at the county level are obtained from the same data sources as the zipcode level data, which are the U.S. Census Bureau and the Federal Housing Finance Agency, respectively. Unemployment is the average of the monthly unemployment rate within each year.

Table 4
Local Characteristics and Second-lien Misrepresentation

	(1)	(2)	(3)	(4)
CPRATIO	0.0024***	0.0020***	0.0018***	0.0014***
	(0.001)	(0.000)	(0.001)	(0.000)
REL	-0.0013***	-0.0012***	-0.0006	-0.0001
	(0.000)	(0.000)	(0.000)	(0.000)
Education	0.0011**	0.0005	0.0018***	0.0005*
	(0.001)	(0.000)	(0.000)	(0.000)
Married	-0.0006	0.0002	-0.0000	0.0002
	(0.001)	(0.000)	(0.001)	(0.000)
Income	-0.0002	-0.0004	-0.0004*	-0.0004**
	(0.000)	(0.000)	(0.000)	(0.000)
Urban	0.0000	-0.0001	-0.0011**	-0.0007**
	(0.001)	(0.000)	(0.000)	(0.000)
Over65	0.0034***	0.0018***	0.0003	0.0006**
	(0.000)	(0.000)	(0.000)	(0.000)
Male-female ratio	0.0006	0.0011***	0.0000	-0.0000
	(0.000)	(0.000)	(0.000)	(0.000)
Minority	-0.0006	0.0001	-0.0018***	0.0006
	(0.001)	(0.001)	(0.001)	(0.000)
Part I crime	0.0001	0.0004	0.0002	0.0008**
	(0.001)	(0.001)	(0.001)	(0.000)
Part II crime	-0.0012**	-0.0005	-0.0005	0.0001
	(0.001)	(0.000)	(0.000)	(0.000)
Democratic	-0.0014	-0.0013	-0.0022*	-0.0009
	(0.001)	(0.001)	(0.001)	(0.001)
Total population	0.0024***	0.0023***	0.0040***	0.0010***
	(0.001)	(0.001)	(0.001)	(0.000)
Unemployment	0.0037***	0.0019***	0.0051***	-0.0008
	(0.001)	(0.001)	(0.001)	(0.001)
HPA	-0.0050***	-0.0009	-0.0035***	0.0017***
	(0.001)	(0.001)	(0.001)	(0.001)
Simul. Second	0.3387***	0.3097***	0.3104***	0.3130***
	(0.002)	(0.002)	(0.002)	(0.002)
Interest rate	-0.0119***	-0.0120***	-0.0112***	-0.0072***
	(0.000)	(0.001)	(0.000)	(0.000)
FICO	-0.0093***	-0.0071***	-0.0072***	-0.0062***
	(0.000)	(0.000)	(0.000)	(0.000)
Balance	0.0011***	0.0030***	0.0027***	0.0048***
	(0.000)	(0.000)	(0.000)	(0.000)
LTV	0.0036***	0.0056***	0.0058***	0.0056***
	(0.000)	(0.000)	(0.000)	(0.000)
ARM	0.0017**	0.0121***	0.0115***	0.0088***
	(0.001)	(0.001)	(0.001)	(0.001)
Negative amortization	-0.0279***	-0.0149***	-0.0128***	-0.0041***
	(0.001)	(0.002)	(0.002)	(0.001)
Option ARM	-0.0023***	-0.0069***	-0.0072***	-0.0080***
	(0.001)	(0.001)	(0.000)	(0.000)
Prepayment penalty	-0.0491***	-0.0434***	-0.0428***	-0.0227***
	(0.003)	(0.005)	(0.005)	(0.005)
Low or no doc.	-0.0088***	-0.0097***	-0.0107***	-0.0082***
	(0.001)	(0.001)	(0.001)	(0.001)
Cash-out	0.0182***	0.0182***	0.0180***	0.0182***
	(0.001)	(0.001)	(0.001)	(0.001)
No-cash-out	0.0144***	0.0151***	0.0153***	0.0158***
	(0.001)	(0.001)	(0.001)	(0.001)
Investment	0.0234***	0.0256***	0.0252***	0.0258***
	(0.001)	(0.001)	(0.001)	(0.001)
Second-home	0.0107***	0.0138***	0.0139***	0.0158***
	(0.001)	(0.001)	(0.001)	(0.001)
Originator FE	N	Y	Y	Y
State FE	N	N	Y	Y
Half-year FE	N	N	N	Y
Observations	2,831,176	2,353,016	2,353,016	2,353,016
Adj. R^2	0.284	0.313	0.313	0.318

The table shows OLS estimates from regressions where the dependent variable takes a value of one if the loan is recognized as second-lien misreported and zero otherwise. Specific fixed effects are used if indicated by Y and not used if indicated by N. Variables are defined in Table 1. All continuous variables are winsorized at the 0.5 percent level and then standardized. Standard errors clustered at the county level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 5
Gambling Preference and Second-lien Misrepresentation - Borrower Incentive

	(1) Primary	(2) Non-primary	(3) Purchase	(4) Refinance
CPRATIO	0.0014***	0.0008	0.0022***	0.0006*
	(0.000)	(0.001)	(0.001)	(0.000)
REL	-0.0001	-0.0003	-0.0005	0.0001
	(0.000)	(0.001)	(0.001)	(0.000)
Education	0.0007**	-0.0011*	0.0008	0.0003
	(0.000)	(0.001)	(0.001)	(0.000)
Married	0.0004	-0.0004	0.0008	-0.0002
	(0.000)	(0.001)	(0.001)	(0.000)
Income	-0.0011***	0.0016***	-0.0003	-0.0003**
	(0.000)	(0.000)	(0.000)	(0.000)
Urban	-0.0008**	0.0010	-0.0008	-0.0006**
	(0.000)	(0.001)	(0.001)	(0.000)
Over65	0.0006**	0.0004	0.0001	0.0005**
	(0.000)	(0.000)	(0.000)	(0.000)
Male-female ratio	-0.0001	0.0007	-0.0001	0.0003
	(0.000)	(0.001)	(0.000)	(0.000)
Minority	0.0007	-0.0006	0.0009	0.0003
	(0.000)	(0.001)	(0.001)	(0.000)
Part I crime	0.0010**	0.0009	0.0016**	0.0003
	(0.000)	(0.001)	(0.001)	(0.000)
Part II crime	0.0001	0.0000	0.0001	0.0000
	(0.000)	(0.001)	(0.001)	(0.000)
Democratic	-0.0009	-0.0005	-0.0012	-0.0005
	(0.001)	(0.001)	(0.001)	(0.000)
Total population	0.0012**	-0.0010	0.0020**	0.0007**
	(0.000)	(0.001)	(0.001)	(0.000)
Unemployment	-0.0008	-0.0011*	-0.0010	-0.0001
	(0.001)	(0.001)	(0.001)	(0.000)
HPA	0.0018***	-0.0000	0.0021**	0.0019***
	(0.001)	(0.001)	(0.001)	(0.000)
Simul. Second	0.3030***	0.4114***	0.2774***	0.3831***
	(0.002)	(0.006)	(0.002)	(0.003)
Interest rate	-0.0087***	-0.0000	-0.0156***	-0.0014***
	(0.000)	(0.001)	(0.001)	(0.000)
FICO	-0.0063***	-0.0042***	-0.0124***	-0.0023***
	(0.000)	(0.000)	(0.001)	(0.000)
Balance	0.0053***	0.0019***	0.0062***	0.0016***
	(0.000)	(0.000)	(0.000)	(0.000)
LTV	0.0050***	0.0090***	0.0172***	0.0009***
	(0.000)	(0.000)	(0.001)	(0.000)
ARM	0.0116***	-0.0153***	0.0191***	0.0009***
	(0.001)	(0.001)	(0.001)	(0.000)
Negative amortization	-0.0092***	0.0316***	-0.0244***	0.0052***
	(0.001)	(0.002)	(0.003)	(0.001)
Option ARM	-0.0093***	-0.0089***	-0.0084***	-0.0040***
	(0.000)	(0.001)	(0.001)	(0.000)
Prepayment penalty	-0.0254***	-0.0110**	-0.0566***	-0.0059***
	(0.006)	(0.005)	(0.010)	(0.003)
Low or no doc.	-0.0077***	-0.0088***	-0.0104***	-0.0048***
	(0.001)	(0.001)	(0.002)	(0.000)
Cash-out	0.0148***	0.0308***		
	(0.001)	(0.001)		
No-cash-out	0.0133***	0.0264***		
	(0.001)	(0.001)		
Investment			0.0255***	0.0192***
			(0.001)	(0.001)
Second-home			0.0171***	0.0101***
			(0.001)	(0.001)
Originator FE	Y	Y	Y	Y
State FE	Y	Y	Y	Y
Half-year FE	Y	Y	Y	Y
Observations	2,052,270	300,095	1,007,280	1,344,733
Adj. R^2	0.312	0.418	0.315	0.382

The table shows OLS estimates from regressions where the dependent variable takes a value of one if the loan is recognized as second-lien misreported and zero otherwise. We look at different subsamples by borrowers incentive structures: primary or non-primary (columns (1) and (2)) and purchase or refinance (columns (3) and (4)). Variables are defined in Table 1. All continuous variables are winsorized at the 0.5 percent level and then standardized. All specifications include state, half-year, and originator fixed effects. Standard errors clustered at the county level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 6
Gambling Preference and Second-Lien Misrepresentation – Credit Expansion and Belief Distortion

	(1) High Income	(2) Low Income	(3) High FICO	(4) Low FICO	(5) High HPA	(6) Low HPA
CPRATIO	0.0013**	0.0017***	0.0012**	0.0017***	0.0019***	0.0008
	(0.001)	(0.001)	(0.001)	(0.000)	(0.000)	(0.001)
REL	0.0001	-0.0010**	-0.0008*	0.0003	-0.0010***	0.0006
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Education	0.0007**	-0.0002	0.0003	0.0010**	0.0010**	0.0006
	(0.000)	(0.001)	(0.000)	(0.000)	(0.000)	(0.000)
Married	-0.0000	0.0008	0.0007	-0.0006	0.0005	0.0002
	(0.000)	(0.001)	(0.001)	(0.000)	(0.000)	(0.001)
Income	0.0001	-0.0017**	-0.0001	-0.0008**	-0.0007**	-0.0001
	(0.000)	(0.001)	(0.000)	(0.000)	(0.000)	(0.000)
Urban	-0.0004	-0.0013***	-0.0003	-0.0011***	-0.0012***	-0.0002
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Over65	0.0007**	0.0002	0.0003	0.0005	0.0003	0.0006*
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Male-female ratio	0.0003	-0.0006*	0.0005	-0.0004	-0.0005	0.0006
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Minority	0.0009	-0.0000	0.0007	0.0004	0.0009*	0.0003
	(0.001)	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)
Part I crime	0.0008	0.0012**	0.0012**	0.0005	0.0002	0.0013**
	(0.000)	(0.001)	(0.001)	(0.000)	(0.000)	(0.001)
Part II crime	-0.0001	0.0006	-0.0000	0.0003	0.0008**	-0.0007
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Democratic	-0.0013*	-0.0001	-0.0007	-0.0010	-0.0027***	0.0015*
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Total population	0.0007	0.0021***	0.0016***	0.0005	0.0019***	0.0006
	(0.000)	(0.001)	(0.001)	(0.001)	(0.000)	(0.001)
Unemployment	-0.0005	-0.0012**	-0.0013**	-0.0003	-0.0004	-0.0004
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
HPA	0.0025***	-0.0004	0.0021***	0.0021***	0.0005	0.0055***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Simul. Second	0.3115***	0.3192***	0.2821***	0.3583***	0.3168***	0.3086***
	(0.002)	(0.003)	(0.003)	(0.002)	(0.003)	(0.002)
Interest rate	-0.0069***	-0.0087***	-0.0026***	-0.0127***	-0.0076***	-0.0068***
	(0.000)	(0.001)	(0.000)	(0.000)	(0.000)	(0.001)
FICO	-0.0055***	-0.0086***	-0.0027***	-0.0140***	-0.0057***	-0.0070***
	(0.000)	(0.001)	(0.000)	(0.001)	(0.000)	(0.001)
Balance	0.0044***	0.0060***	0.0047***	0.0032***	0.0050***	0.0043***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
LTV	0.0052***	0.0076***	0.0026***	0.0132***	0.0055***	0.0058***
	(0.000)	(0.001)	(0.000)	(0.001)	(0.000)	(0.000)
ARM	0.0092***	0.0067***	0.0093***	0.0024***	0.0068***	0.0112***
	(0.001)	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)
Negative amortization	-0.0039***	-0.0039**	0.0071***	-0.0086***	-0.0034**	-0.0055***
	(0.001)	(0.002)	(0.001)	(0.001)	(0.001)	(0.002)
Option ARM	-0.0078***	-0.0085***	-0.0065***	-0.0017**	-0.0079***	-0.0080***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Prepayment penalty	-0.0255***	-0.0146**	-0.0305***	-0.0086*	-0.0279***	-0.0138**
	(0.005)	(0.007)	(0.005)	(0.005)	(0.006)	(0.007)
Low or no doc.	-0.0084***	-0.0085***	-0.0127***	-0.0112***	-0.0089***	-0.0071***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Cash-out	0.0185***	0.0179***	0.0238***	0.0176***	0.0188***	0.0177***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
No-cash-out	0.0158***	0.0162***	0.0166***	0.0148***	0.0158***	0.0157***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Investment	0.0256***	0.0267***	0.0235***	0.0317***	0.0251***	0.0266***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Second-home	0.0159***	0.0136***	0.0150***	0.0197***	0.0155***	0.0161***
	(0.001)	(0.002)	(0.001)	(0.002)	(0.001)	(0.001)
Fixed Effects	Y	Y	Y	Y	Y	Y
Observations	1,809,909	542,366	1,308,243	1,044,300	1,352,931	999,126
Adj. R^2	0.315	0.329	0.298	0.358	0.321	0.315

The table shows OLS estimates from regressions where the dependent variable takes a value of one if the loan is recognized as second-lien misreported and zero otherwise. We look at different subsamples: high or low income ((1) and (2)), high or low credit score ((3) and (4)), and high or low HPA areas ((5) and (6)). The high or low income subsamples are defined by comparing local household median income with the sample median at the ZIP code level. The high or low credit score subsamples are divided by a FICO score of 670. The high and low HPA subsamples are defined based on the state-year median HPA, calculated from all available ZIP code-level data. All continuous variables are winsorized at the 0.5 percent level and then standardized. All specifications include state, half-year, and originator fixed effects. Standard errors clustered at the county level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 7
Discontinuities for Low-documentation Loans Around the Credit Threshold in All Areas

	(1) All loan	(2) Reported second	(3) Misreported second
Panel A. Jump Ratio in Number of Loans			
Est. 620+/620-	2.266	4.939	3.838
Panel B. Default Rate			
FICO$\geq$620 (β)	0.039	0.045	0.046
t-stat	(3.92)	(2.17)	(-0.84)

The table reports the discontinuities for low-documentation loans around a FICO score of 620 in all areas. Panel A presents the estimates from regressions where the dependent variable is the number of loans at each FICO score. Using local linear regressions of the RDD approach, we estimate the number of loans at FICO 620^- and FICO 620^+ and compute the ratio of the estimated number of loans at FICO 620^+ to 620^-. Panel B presents the estimates from regressions where the dependent variable takes a value of one if the loan becomes 90 days or more delinquent using the MBA method in the first three years after origination and zero otherwise. Using local linear regressions of the RDD approach with control variables, we estimate the difference in default rates between FICO 620^- and FICO 620^+. We perform the estimation for all loans, loans with correctly reported simultaneous seconds, and loans with misreported simultaneous seconds, and the results are reported in columns (1) to (3), respectively. Bias-corrected t-values (standard errors clustered at the county level) following Calonico et al. (2014) are reported in parentheses.

Table 8
Discontinuities for Low-documentation Loans Around the Credit Threshold in High and Low Gambling Preference Areas

	(1) High	(2) Low	(3) High-Low
Panel A. Jump Ratio in Number of Loans (Est. 620+/620-)			
All loan	2.240	2.289	-0.049
			(-0.52)
Reported second	5.498	4.581	0.917
			(1.23)
Misreported second	2.763	4.826	-2.063
			(-2.83)
Panel B. Default Rate			
All loan	0.039	0.038	0.001
	(3.44)	(2.36)	(0.07)
Reported second	0.011	0.065	-0.053
	(0.22)	(2.65)	(-1.38)
Misreported second	0.016	0.077	-0.061
	(0.13)	(-1.09)	(-1.33)

The table reports the discontinuities for low-documentation loans around a FICO score of 620 separately in high and low gambling preference areas. Panel A presents the estimates from regressions where the dependent variable is the number of loans at each FICO score. Using local linear regressions of the RDD approach, we estimate the number of loans at FICO 620^- and FICO 620^+ and compute the ratio of the estimated number of loans at FICO 620^+ to 620^-. Panel B presents the estimates from regressions where the dependent variable is the default dummy variable. Using local linear regressions of the RDD approach with control variables, we estimate the difference in default rates between FICO 620^- and FICO 620^+. Using the diff-in-disc approach, we estimate the difference between high and low gambling preference areas. We perform the estimation for all loans, loans with correctly reported simultaneous seconds, and loans with misreported simultaneous seconds. The results for loans in high, low, and the difference between high and low gambling preference areas are reported in columns (1) to (3), respectively. For discontinuities, bias-corrected t-values (standard errors clustered at the county level) following Calonico et al. (2014) are reported in parentheses. For the difference between high and low gambling preference areas, t-statistics based on heteroskedasticity-consistent standard errors are reported in parentheses.

Table 9
Outcomes of Gambling Preference-Related Misrepresentation

	(1)	(2)	(3)
Misreported second	0.120***	0.120***	0.121***
	(0.004)	(0.004)	(0.003)
CPRATIO		-0.000	-0.001
		(0.003)	(0.003)
Misreported second*CPRATIO			0.014***
			(0.002)
REL	-0.008***	-0.008***	-0.008***
	(0.003)	(0.003)	(0.003)
Education	0.005*	0.005*	0.005*
	(0.003)	(0.002)	(0.002)
Married	0.020***	0.020***	0.020***
	(0.003)	(0.003)	(0.003)
Income	-0.012***	-0.012***	-0.012***
	(0.002)	(0.002)	(0.002)
Urban	0.003	0.003	0.003
	(0.003)	(0.003)	(0.003)
Over65	0.003	0.003	0.003
	(0.003)	(0.003)	(0.003)
Male-female ratio	-0.001	-0.001	-0.001
	(0.002)	(0.002)	(0.002)
Minority	0.002	0.002	0.002
	(0.004)	(0.004)	(0.004)
Part I crime	0.014***	0.014***	0.014***
	(0.004)	(0.004)	(0.004)
Part II crime	-0.005**	-0.005**	-0.005**
	(0.002)	(0.002)	(0.002)
Democratic	0.014***	0.014***	0.014***
	(0.005)	(0.005)	(0.005)
Total population	-0.000	-0.000	-0.000
	(0.005)	(0.005)	(0.005)
Unemployment	0.004	0.004	0.004
	(0.003)	(0.003)	(0.003)
HPA	0.048***	0.048***	0.048***
	(0.004)	(0.004)	(0.004)
Reported second	0.142***	0.142***	0.142***
	(0.007)	(0.007)	(0.007)
Interest rate	0.033***	0.033***	0.033***
	(0.001)	(0.001)	(0.001)
FICO	-0.093***	-0.093***	-0.094***
	(0.002)	(0.002)	(0.002)
Balance	0.023***	0.023***	0.023***
	(0.002)	(0.002)	(0.002)
LTV	0.052***	0.052***	0.052***
	(0.002)	(0.002)	(0.002)
ARM	0.041***	0.041***	0.041***
	(0.002)	(0.002)	(0.002)
Negative amortization	0.037***	0.037***	0.037***
	(0.004)	(0.004)	(0.004)
Option ARM	0.054***	0.054***	0.054***
	(0.002)	(0.002)	(0.002)
Prepayment penalty	0.047***	0.047***	0.048***
	(0.007)	(0.007)	(0.007)
Low or no doc.	0.055***	0.055***	0.055***
	(0.002)	(0.002)	(0.002)
Cash-out	-0.025***	-0.025***	-0.025***
	(0.002)	(0.002)	(0.002)
No-cash-out	0.010***	0.010***	0.010***
	(0.004)	(0.004)	(0.004)
Investment	0.045***	0.045***	0.045***
	(0.004)	(0.004)	(0.004)
Second-home	0.008**	0.008**	0.008**
	(0.003)	(0.003)	(0.003)
Fixed Effects	Y	Y	Y
Observations	2,351,791	2,351,791	2,351,791
Adj. R^2	0.226	0.226	0.226

The table shows OLS estimates from regressions where the dependent variable takes a value of one if the loan becomes 90 days or more delinquent using the MBA method in the first three years after origination and zero otherwise. All continuous variables are winsorized at the 0.5 percent level and then standardized. All specifications include state, half-year, and originator fixed effects. Standard errors clustered at the county level are reported in parentheses. $^*p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 10
Gambling Preference and Mortgage Misrepresentation - Probit Model in Simultaneous Seconds Sample

	(1)	(2)	(3)	(4)
CPRATIO	0.014***	0.009***	0.009***	0.005***
	(0.003)	(0.002)	(0.003)	(0.002)
REL	-0.006***	-0.004***	-0.003*	-0.001
	(0.002)	(0.002)	(0.002)	(0.001)
Education	0.006**	0.003	0.011***	0.002*
	(0.003)	(0.002)	(0.002)	(0.001)
Married	0.001	0.006***	-0.000	0.000
	(0.003)	(0.002)	(0.002)	(0.001)
Income	-0.002	-0.001	-0.002*	-0.002*
	(0.001)	(0.001)	(0.001)	(0.001)
Urban	-0.000	0.001	-0.006***	-0.003*
	(0.003)	(0.002)	(0.002)	(0.001)
Over65	0.017***	0.008***	0.000	0.001
	(0.003)	(0.002)	(0.002)	(0.001)
Male-female ratio	0.004	0.005***	0.000	-0.000
	(0.002)	(0.002)	(0.002)	(0.001)
Minority	0.000	0.005**	-0.009***	0.003*
	(0.004)	(0.003)	(0.003)	(0.002)
Part I crime	0.001	0.003	0.001	0.002
	(0.004)	(0.002)	(0.003)	(0.002)
Part II crime	-0.006**	-0.002	-0.003	-0.001
	(0.003)	(0.002)	(0.002)	(0.001)
Democratic	-0.007	-0.008**	-0.012**	-0.003
	(0.006)	(0.004)	(0.005)	(0.002)
Total population	0.012***	0.009***	0.019***	0.005***
	(0.004)	(0.002)	(0.003)	(0.002)
Unemployment	0.017***	0.011***	0.032***	0.001
	(0.003)	(0.003)	(0.004)	(0.002)
HPA	-0.017***	-0.010***	-0.024***	0.004**
	(0.004)	(0.003)	(0.003)	(0.002)
Loan-level controls	Y	Y	Y	Y
Originator FE	N	Y	Y	Y
State FE	N	N	Y	Y
Half-year FE	N	N	N	Y
Observations	590,726	472,290	460,962	441,794

The table shows the marginal effects of CPRATIO from probit models where the dependent variable takes a value of one if the loan is misreported and zero otherwise. The sample is restricted to the loans with simultaneous second liens. All continuous variables, except for the crime rates, are winsorized at the 0.5 percent level and then standardized. All specifications include state, half-year, and originator fixed effects. Standard errors clustered at the county level are reported in parentheses. *$p < 0.10$, **$p < 0.05$, and ***$p < 0.01$.

Table 11
Gambling Preference and Mortgage Misrepresentation - Probit Model in Whole Sample

	(1)	(2)	(3)	(4)
CPRATIO	0.0033***	0.0029***	0.0015**	0.0015**
	(0.001)	(0.001)	(0.001)	(0.001)
REL	-0.0017***	-0.0011***	-0.0010**	-0.0005
	(0.000)	(0.000)	(0.000)	(0.000)
Education	0.0029***	0.0015**	0.0007	0.0002
	(0.001)	(0.001)	(0.001)	(0.001)
Married	0.0022**	0.0020**	0.0025***	0.0023***
	(0.001)	(0.001)	(0.001)	(0.001)
Income	-0.0004	0.0002	-0.0001	-0.0002
	(0.001)	(0.000)	(0.000)	(0.000)
Urban	-0.0007	-0.0004	-0.0005	-0.0004
	(0.001)	(0.000)	(0.000)	(0.000)
Over65	-0.0013**	-0.0014***	-0.0007*	0.0000
	(0.001)	(0.000)	(0.000)	(0.000)
Male-female ratio	-0.0001	0.0003	0.0000	0.0000
	(0.001)	(0.000)	(0.000)	(0.000)
Minority	-0.0016	-0.0004	-0.0010	0.0005
	(0.001)	(0.001)	(0.001)	(0.001)
Part I crime	0.0013	0.0008	0.0015**	0.0017***
	(0.001)	(0.001)	(0.001)	(0.001)
Part II crime	0.0006	0.0010**	-0.0000	0.0004
	(0.001)	(0.000)	(0.000)	(0.000)
Democratic	0.0011	0.0011	0.0000	0.0007
	(0.001)	(0.001)	(0.001)	(0.001)
Total population	0.0017	0.0012	0.0026***	0.0007
	(0.001)	(0.001)	(0.001)	(0.000)
Unemployment	0.0030***	0.0013***	0.0010	-0.0026***
	(0.001)	(0.000)	(0.001)	(0.001)
HPA	0.0037***	0.0032***	0.0011	0.0040***
	(0.001)	(0.001)	(0.001)	(0.001)
Loan-level controls	Y	Y	Y	Y
Originator FE	N	Y	Y	Y
State FE	N	N	Y	Y
Half-year FE	N	N	N	Y
Observations	2,831,176	2,336,155	2,305,257	2,247,430

The table shows the marginal effects of CPRATIO from probit models where the dependent variable takes a value of one if the loan is misreported and zero otherwise. All continuous variables, except for the crime rates, are winsorized at the 0.5 percent level and then standardized. All specifications include state, half-year, and originator fixed effects. Standard errors clustered at the county level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 12
Gambling Preference and Mortgage Misrepresentation - Causal Forest

	(1) Misreported second
CPRATIO	0.0017**
	(0.0007)

The table shows causal forest estimates from regressions where the dependent variable takes a value of one if the loan is misreported and zero otherwise. All control variables in Table 4 are used for growing trees and forests. Before growing the trees, all continuous variables are winsorized at the 0.5 percent level and then standardized. A half-year variable is created, starting from the first half of 2005 as 1, and used as a variable in growing trees and forests. Observations are clustered by state, and each unit is given the same weight (so that larger clusters receive more weight). 2000 trees are grown in the causal forest. The fraction used for determining splits (honesty) is 50 percent. Standard errors are reported in parentheses. *$p < 0.10$, **$p < 0.05$, and ***$p < 0.01$.

Table 13
Alternative Gambling Preference Measures

	(1)	(2)
Gambling Industry Employment	0.003***	0.003***
	(0.001)	(0.001)
Casino Hotel Employment		-0.001
		(0.002)
Nonemployer Gambling Establishment		0.001
		(0.002)
REL	0.003	0.002
	(0.002)	(0.002)
Education	-0.000	-0.001
	(0.002)	(0.002)
Married	0.000	0.000
	(0.001)	(0.001)
Income	0.001	0.001
	(0.002)	(0.002)
Urban	0.001	0.001
	(0.002)	(0.002)
Over65	-0.000	-0.000
	(0.001)	(0.001)
Male-female ratio	0.002	0.002
	(0.001)	(0.002)
Minority	-0.003*	-0.003*
	(0.001)	(0.002)
Part I crime	0.004**	0.004**
	(0.002)	(0.002)
Part II crime	0.000	0.000
	(0.002)	(0.002)
Democratic	0.002	0.001
	(0.002)	(0.002)
Total population	0.001	0.001
	(0.002)	(0.002)
Unemployment	-0.001	-0.001
	(0.001)	(0.001)
HPA	0.002	0.002
	(0.002)	(0.002)
Simul. Second	0.316***	0.316***
	(0.004)	(0.004)
Interest rate	-0.008***	-0.008***
	(0.001)	(0.001)
FICO	-0.006***	-0.006***
	(0.001)	(0.001)
Balance	0.005***	0.005***
	(0.000)	(0.000)
LTV	0.006***	0.006***
	(0.000)	(0.000)
ARM	0.008***	0.008***
	(0.001)	(0.001)
Negative amortization	-0.001	-0.001
	(0.002)	(0.002)
Option ARM	-0.022***	-0.022***
	(0.005)	(0.005)
Prepayment penalty	-0.009***	-0.009***
	(0.002)	(0.002)
Low or no doc.	-0.008***	-0.008***
	(0.001)	(0.001)
Cash-out	0.019***	0.019***
	(0.001)	(0.001)
No-cash-out	0.016***	0.016***
	(0.001)	(0.001)
Investment	0.027***	0.027***
	(0.001)	(0.001)
Second-home	0.017***	0.017***
	(0.001)	(0.001)
Originator FE	Y	Y
State FE	Y	Y
Half-year FE	Y	Y
Observations	1,032,077	1,032,077
Adj. R^2	0.318	0.318

The table shows OLS estimates from regressions where the dependent variable takes a value of one if the loan is recognized as second-lien misreported and zero otherwise. The local controls are aggregated to CBSA level. All continuous variables are winsorized at the 0.5 percent level and then standardized. Standard errors clustered at the CBSA level are reported in parentheses. $*p < 0.10$, $**p < 0.05$, and $***p < 0.01$.

Table 14
Local Characteristics and Loan Collateral Fundamentals around Discontinuity in Low-Documentation Loans

	FICO≥620 (β))	t-stat
CPRATIO	-0.041	(-0.035)
REL	0.007	(0.741)
Education (%)	0.619	(1.561)
Married (%)	0.191	(0.135)
Income	0.024	(1.460)
Urban (%)	-0.278	(0.330)
Over65 (%)	-0.386	(-0.883)
Male-female ratio	0.001	(0.147)
Minority (%)	-0.032	(0.123)
Part I crime	-5.507	(-0.418)
Part II crime	14.228	(0.512)
Democratic	-0.011	(-0.049)
Total population (ln)	-0.045	(0.038)
Unemployment (%)	0.024	(-0.210)
HPA	-0.002	(-0.102)
Balance	0.045	(0.868)
LTV (%)	0.637	(-0.329)

The table reports estimates from regressions of local characteristics and loan collateral fundamentals around a FICO score of 620 for low-documentation loans. Local linear regressions using the RDD approach are employed to estimate the differences in these predetermined variables between FICO 620^{-} and FICO 620^{+}. The bias-corrected t-values (standard errors clustered at the county level) following Calonico et al. (2014) are reported in parentheses.

Table 15
Discontinuities in Reported and Misreported Second-Lien Rates around the Credit Threshold in Low-Documentation Loans

		Reported second				Misreported second			
		High	Low	All	High-Low	High	Low	All	High-Low
Raw	FICO$\geq$620 (β)	0.094	0.087	0.090	0.006	0.014	0.065	0.042	-0.051
	t-stat	(7.38)	(8.58)	(11.22)	(0.57)	(0.69)	(9.13)	(6.49)	-(5.19)
Control	FICO$\geq$620 (β)	0.087	0.069	0.078	0.018	0.009	0.052	0.032	-0.043
	t-stat	(7.44)	(8.81)	(11.15)	(1.86)	(0.26)	(7.85)	(5.51)	-(4.48)
Control+FEs	FICO$\geq$620 (β)	0.065	0.058	0.062	0.008	0.006	0.015	0.010	-0.008
	t-stat	(5.85)	(6.23)	(8.35)	(0.89)	(0.97)	(3.33)	(2.97)	-(1.14)

The table reports discontinuities in reported and misreported second-lien rates around a FICO score of 620 in low-documentation loans. Local linear regressions from the RDD approach are used to estimate the difference in rates between FICO 620^- and FICO 620^+, where the dependent variable is either the dummy for correctly reported seconds or the dummy for misreported seconds. Using the diff-in-disc approach, we further estimate the difference between high and low gambling preference areas. Estimations are conducted without controls, with controls, and with both controls and the fixed effects from the baseline regressions. For discontinuities, bias-corrected t-values (standard errors clustered at the county level) following Calonico et al. (2014) are reported in parentheses. For the difference between high and low gambling preference areas, t-statistics based on heteroskedasticity-consistent standard errors are reported in parentheses.

Table 16
Outcomes of Gambling Preference-Related Misrepresentation - Subsamples

	(1) High Income	(2) Low Income	(3) High FICO	(4) Low FICO	(5) High HPA	(6) Low HPA
Misreported second	0.118***	0.131***	0.094***	0.155***	0.128***	0.109***
	(0.003)	(0.004)	(0.003)	(0.004)	(0.004)	(0.004)
CPRATIO	0.002	-0.011**	-0.000	-0.005	0.001	0.004
	(0.003)	(0.005)	(0.003)	(0.003)	(0.005)	(0.005)
Misreported second*CPRATIO	0.013***	0.019***	0.015***	0.015***	0.016***	0.011***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.004)
REL	-0.009***	-0.004	-0.008***	-0.006**	-0.007**	-0.011***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Education	0.007***	0.002	0.004*	0.006**	0.004	0.004
	(0.002)	(0.004)	(0.002)	(0.003)	(0.004)	(0.003)
Married	0.022***	0.020***	0.022***	0.015***	0.021***	0.019***
	(0.003)	(0.004)	(0.003)	(0.003)	(0.005)	(0.004)
Income	-0.013***	-0.004	-0.013***	-0.010***	-0.012***	-0.016***
	(0.002)	(0.003)	(0.001)	(0.003)	(0.003)	(0.002)
Urban	0.002	0.004	0.005	0.002	0.003	0.002
	(0.003)	(0.004)	(0.003)	(0.003)	(0.003)	(0.004)
Over65	0.004	-0.000	-0.000	0.006**	0.001	0.005
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Male-female ratio	-0.001	-0.003	-0.001	-0.001	-0.002	-0.003
	(0.003)	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)
Minority	0.001	0.003	0.002	0.002	-0.000	0.009**
	(0.004)	(0.005)	(0.003)	(0.004)	(0.005)	(0.004)
Part I crime	0.015***	0.014***	0.015***	0.012***	0.016***	0.014***
	(0.003)	(0.005)	(0.003)	(0.004)	(0.004)	(0.004)
Part II crime	-0.005*	-0.006*	-0.004*	-0.005*	-0.006*	-0.004*
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.002)
Democratic	0.012**	0.021***	0.011**	0.015***	0.014**	0.010
	(0.005)	(0.008)	(0.005)	(0.006)	(0.007)	(0.007)
Total population	0.001	-0.001	-0.000	-0.001	-0.006	0.004
	(0.005)	(0.007)	(0.005)	(0.006)	(0.006)	(0.007)
Unemployment	0.006**	0.003	0.007**	0.000	-0.000	0.003
	(0.003)	(0.003)	(0.003)	(0.003)	(0.004)	(0.003)
HPA	0.049***	0.051***	0.045***	0.056***	0.055***	0.054***
	(0.004)	(0.006)	(0.003)	(0.005)	(0.004)	(0.008)
Reported second	0.138***	0.156***	0.117***	0.193***	0.160***	0.117***
	(0.006)	(0.009)	(0.007)	(0.007)	(0.008)	(0.005)
	(0.003)	(0.007)	(0.003)	(0.005)	(0.005)	(0.003)
Loan-level controls	Y	Y	Y	Y	Y	Y
Originator FE	Y	Y	Y	Y	Y	Y
State FE	Y	Y	Y	Y	Y	Y
Half-year FE	Y	Y	Y	Y	Y	Y
Observations	1,809,101	541,951	1,308,003	1,043,316	1,352,234	998,600
Adj. R^2	0.226	0.216	0.225	0.182	0.238	0.209

The table shows OLS estimates from regressions where the dependent variable takes a value of one if the loan is recognized as second-lien misreported and zero otherwise. We look at different subsamples: high or low income ((1) and (2)), high or low credit score ((3) and (4)), and high or low HPA areas ((5) and (6)). The high or low income subsamples are defined by comparing local household median income with the sample median at the ZIP code level. The high or low credit score subsamples are divided by a FICO score of 670. The high and low HPA subsamples are defined based on the state-year median HPA, calculated from all available ZIP code-level data. All continuous variables are winsorized at the 0.5 percent level and then standardized. All specifications include state, half-year, and originator fixed effects. Standard errors clustered at the county level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 17
Alternative Default Measures for Loan Performance Tests

	(1) 90+ delinq in 3 year	(2) 90+ delinq in 2 year	(3) 60+ delinq in 3 year	(4) B/F/REO in 3 year
Misreported second	0.121***	0.066***	0.123***	0.109***
	(0.003)	(0.003)	(0.003)	(0.003)
CPRATIO	-0.001	-0.001	-0.001	-0.002
	(0.003)	(0.002)	(0.003)	(0.003)
Misreported second*CPRATIO	0.014***	0.007***	0.014***	0.012***
	(0.002)	(0.002)	(0.003)	(0.003)
REL	-0.008***	-0.005***	-0.008***	-0.006***
	(0.003)	(0.002)	(0.003)	(0.002)
Education	0.005*	0.003**	0.004	0.004**
	(0.002)	(0.002)	(0.002)	(0.002)
Married	0.020***	0.011***	0.020***	0.016***
	(0.003)	(0.002)	(0.003)	(0.002)
Income	-0.012***	-0.009***	-0.012***	-0.009***
	(0.002)	(0.001)	(0.002)	(0.002)
Urban	0.003	0.002	0.003	0.003
	(0.003)	(0.002)	(0.003)	(0.003)
Over65	0.003	0.003	0.002	0.004
	(0.003)	(0.002)	(0.003)	(0.003)
Male-female ratio	-0.001	-0.001	-0.002	-0.001
	(0.002)	(0.001)	(0.002)	(0.002)
Minority	0.002	0.002	0.002	-0.001
	(0.004)	(0.002)	(0.004)	(0.003)
Part I crime	0.014***	0.008***	0.013***	0.014***
	(0.004)	(0.002)	(0.004)	(0.003)
Part II crime	-0.005**	-0.002	-0.005**	-0.005**
	(0.002)	(0.002)	(0.003)	(0.002)
Democratic	0.014***	0.006*	0.016***	0.012**
	(0.005)	(0.003)	(0.005)	(0.005)
Total population	-0.000	0.001	-0.000	-0.002
	(0.005)	(0.003)	(0.005)	(0.004)
Unemployment	0.004	0.005**	0.003	0.005*
	(0.003)	(0.002)	(0.003)	(0.003)
HPA	0.048***	0.026***	0.048***	0.039***
	(0.004)	(0.003)	(0.004)	(0.003)
Reported second	0.142***	0.079***	0.145***	0.114***
	(0.007)	(0.005)	(0.007)	(0.006)
Loan-level controls	Y	Y	Y	Y
Originator FE	Y	Y	Y	Y
State FE	Y	Y	Y	Y
Half-year FE	Y	Y	Y	Y
Observations	2,351,791	2,351,791	2,351,791	2,351,791
Adj. R^2	0.226	0.157	0.237	0.156

The table shows OLS estimates from regressions in which the dependent variable uses different measures of default. Column (1) uses 90 days or more delinquency using MBA method in the first three years after origination (baseline). Column (2) uses 90 days or more delinquency using MBA method in the first two years after origination. Column (3) uses 60 days or more delinquency using MBA method in the first two years after origination. Column (4) uses bankruptcy/foreclosure/REO in the first three years after origination. All continuous variables are winsorized at 0.5 percent level and then standardized. All specifications include state, half-year, and originator fixed effects, and standard errors clustered at the county level are reported in the parentheses. $*p < 0.10$, $**p < 0.05$, and $***p < 0.01$.

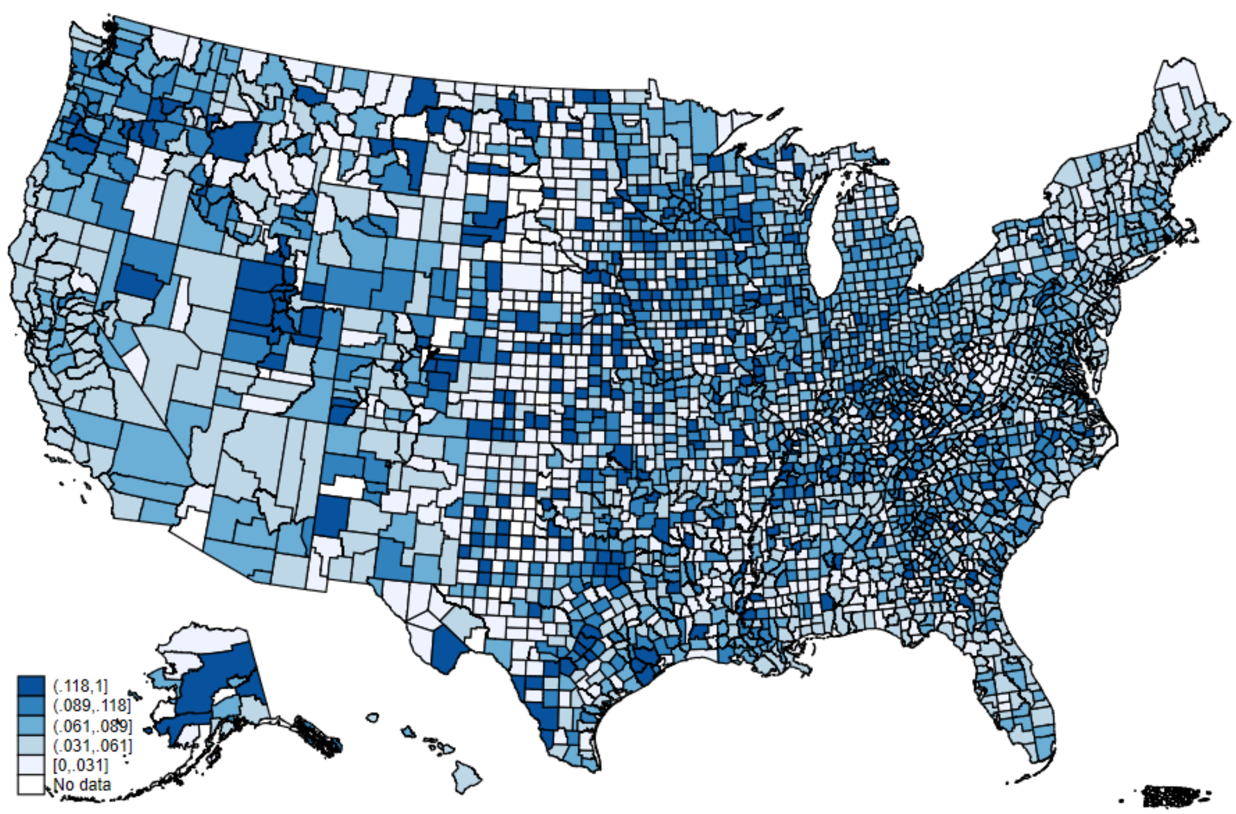

Figure 1
Second-lien Misrepresentations Distribution
The figure plots the county level proportion of second-lien misrepresentation in all loans.

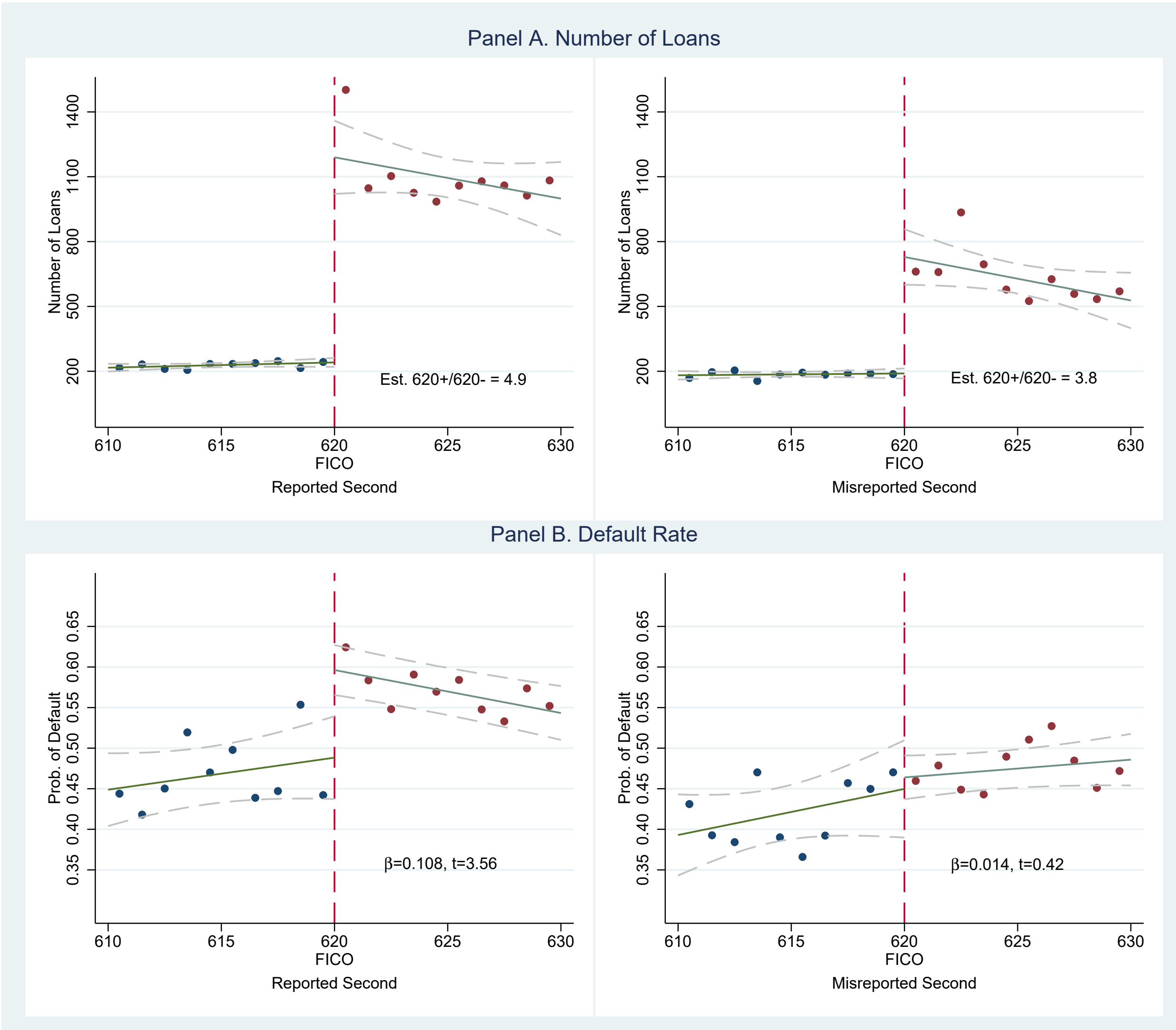

Figure 2

Discontinuities for Low-Documentation Loans with Simultaneous Seconds Around Credit Score Thresholds

The figure plots discontinuities for low-documentation loans with simultaneous seconds by credit score. Panel A reports the number of loans, and Panel B reports the default rate. The circles represent the average number of loans or probability of default for each credit score between 610 and 629. The solid lines show the linear fit on either side of the credit score threshold, with dashed lines denoting the 95% confidence interval. The left subfigures report results for loans with correctly reported simultaneous seconds, and the right subfigures present results for loans with misreported simultaneous seconds. As shown, the jump ratio in the number of loans is smaller for misreported seconds than for correctly reported seconds. Additionally, the jump in default rates is large and significant for correctly reported seconds but small and insignificant for misreported seconds.

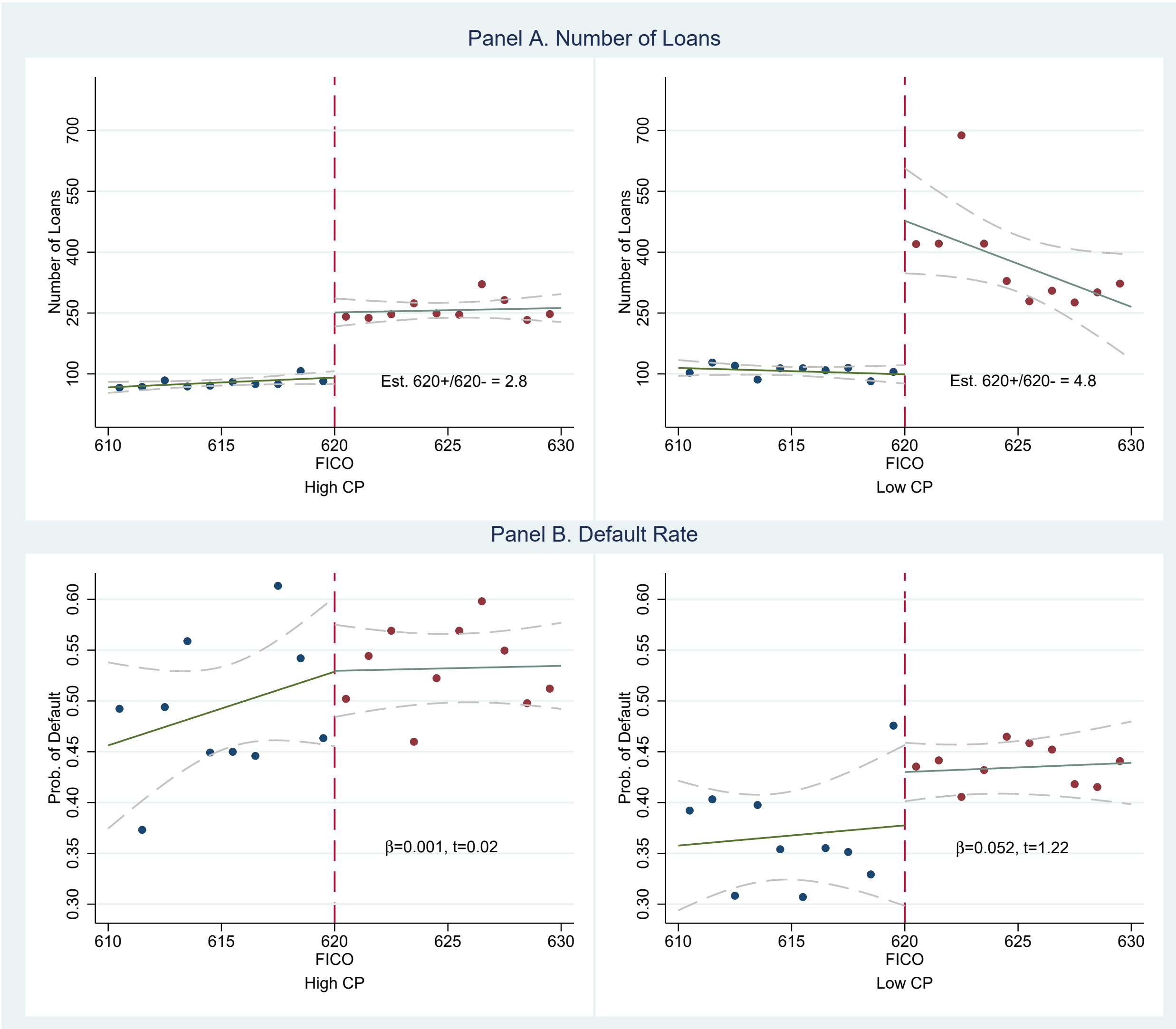

Figure 3

Discontinuities for Low-Documentation Loans with Misreported Simultaneous Seconds Around Credit Score Thresholds by Gambling Preference

The figure plots discontinuities for low-documentation loans with misreported simultaneous seconds by credit score. Panel A reports the number of loans, and Panel B reports the default rate. The circles represent the average number of loans for each credit score between 610 and 629. The solid lines show the linear fit on either side of the credit score threshold, with dashed lines denoting the 95% confidence interval. The left subfigures report results for loans in high gambling preference areas, while the right subfigures present results for loans in low gambling preference areas. As shown, the jump ratio is much smaller in high gambling preference areas than in low gambling preference areas. Additionally, the jump in default rates is insignificant in both groups, though larger in magnitude in low gambling preference areas.

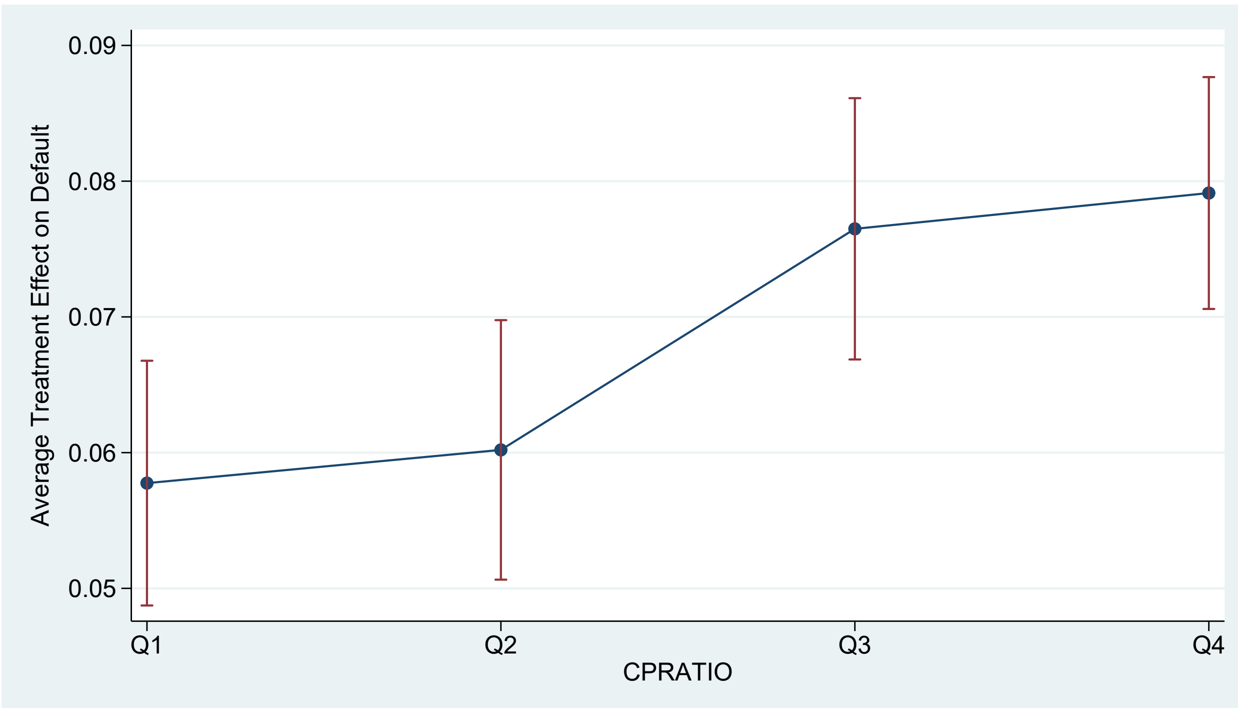

Figure 4
Heterogeneous Treatment Effects of Second-lien Misrepresentation on Default Conditioned on Different Levels of Gambling Preference
The figure plots the heterogeneous treatment effects of second-lien misrepresentation on default, conditioned on different levels of gambling preference, using the causal forest approach. The dependent variable is an indicator that takes a value of one if the loan becomes 90 days or more delinquent using the MBA method in the first three years after origination. The average treatment effect of mortgage misrepresentation is estimated at different levels of local gambling preference (four quarters divided by CPRATIO). All control variables in column (3) of Table 9 are used for growing trees and forests. Before growing the trees, all continuous variables are winsorized at the 0.5 percent level and then standardized. A half-year variable is created, starting from the first half of 2005 as 1, and used as a variable in growing trees and forests. Observations are clustered by state, and each unit is given the same weight (so that larger clusters receive more weight). 2000 trees are grown in the causal forest. The fraction used for determining splits (honesty) is 50 percent.

Appendix

A Alternative Gambling Preference Measures

In this section, we provide detailed information regarding the use of alternative gambling preference measures.

A.1 Data and Measure Construction

Since CPRATIO may capture broader religious influences, such as attitudes toward forgiveness and dishonesty, we also employ more direct measures of local gambling preference. Specifically, we use per capita employment in the gambling industry to quantify local gambling engagement.

To construct the measure, we extract employment data at the core-based statistical area (CBSA) level[1] for the years 2005 through 2007 from the County Business Patterns (CBP) dataset provided by the U.S. Census Bureau. To identify gambling-specific industries, we retain observations with NAICS code 7132, which includes 71321 for Casinos (except Casino Hotels) and 71329 for Other Gambling Industries. These counts are then normalized by dividing by the total population of the area. Population figures for 2005 to 2007 are linearly interpolated based on CBSA-level data from the 2000 and 2010 Census.

In addition to the gambling preference measure described above, we construct supplementary proxies. Because CBP classifies businesses based on their primary services, casino hotels (NAICS code 72110) are excluded from the gambling industry category. However, as these businesses also provide gambling services to the public, we extract employment data for casino hotels separately, normalizing it by total population as well.

Since the CBP dataset includes only employer-based businesses, some small gambling venues without formal employees may be omitted. To account for these, we incorporate data from the Nonemployer Statistics (NES) datasets, also maintained by the U.S. Census Bureau. We retain NES observations with NAICS code 7132[2]. The number of nonemployer gambling establishments is likewise normalized by CBSA population.

While the CBP dataset records the number of establishments without restrictions, the employment data suffers from a key limitation: many cells are suppressed to preserve con-

[1]This refers to the MSA File in the CBP dataset; however, the MSA identification codes correspond to CBSA codes.

[2]During our sample period, no casino hotel reports having employees; therefore, NAICS code 72110 does not appear in the NES data.

fidentiality (Eckert et al., 2021)[3]. To ensure sufficient CBSA-level coverage, we apply the following method to approximate employment figures[4]. First, for each CBSA in which a gambling-related NAICS code appears, if the employment value is nonzero, we retain it as-is. Second, for cases where the employment value is reported as zero, we estimate it by multiplying the number of establishments in each employment-size interval by the median value of that interval. For instance, in CBSA 11460, there are two establishments, one in the 1–4 interval and one in the 5–9 interval, so the estimated employment is calculated as $1 \times 2.5 + 1 \times 7 = 9.5$. For establishments falling within the highest interval (i.e., open-ended, 5,000 or more), we conservatively assign the lower bound value of 5,000. Third, we compare the estimated employment figure with the lower bound of the corresponding employment level flag and retain the value that is greater[5].

The NES dataset also contains limitations: numerous cells are suppressed to preserve confidentiality or to avoid quality issues. In such cases, the number of establishments is recorded as zero; however, the establishment flags indicate that the zero does not reflect true nonexistence. Instead, these suppressed values result from either unwillingness to disclose individual business information or concerns regarding data reliability. To address this issue, we implement the following imputation strategy. For observations with flag `D` (confidentiality concerns), we replace the zero with a value of 2[6]. For observations with flag `S` (reliability concerns), we substitute the zero with the state-level average for the corresponding industry category.

For geographic control variables, the U.S. Census Bureau does not provide CBSA-level data for the year 2000. To address this limitation, we aggregate county-level data up to the CBSA level. For ratio variables, we first aggregate the numerator and denominator separately across constituent counties, and then compute the ratio based on the aggregated values. For unemployment rates obtained from the U.S. Bureau of Labor Statistics, we apply the same aggregation procedure to the annual average data and implement a one-year lag in our regressions. For HPA, we use the CBSA-level HPI data directly from the FHFA.

[3] Another limitation discussed in Eckert et al. (2021) is the periodic reclassification of industries. However, since changes to gambling-related NAICS codes are infrequent and our sample period is short, this concern is minimal for our analysis.

[4] Eckert et al. (2021) proposes a solution for county-level data, but not for CBSA-level data. We adapt their core approach in a simplified form due to constraints that are difficult to satisfy at the CBSA level. Our primary strategy involves using the median value within employment intervals and verifying that the estimated employment remains within the bounds of the corresponding employment level flag interval.

[5] Refer to Tables 1 and 3 in Eckert et al. (2021) for definitions of employment intervals and employment level flag intervals, respectively. In our sample, all estimated values fall below the corresponding upper bounds of the employment level flags.

[6] We use 2 since the smallest observable nonzero value is 3. In untabulated results, substituting with 1 does not materially affect our findings.

Table A1 presents the summary statistics for all constructed gambling preference measures.

A.2 Additional Tests

To validate our alternative measures and explore their relation with CPRATIO, we do the following tests.

First, to examine the relationship between CPRATIO and both the alternative gambling preference measure and the two additional control variables, we conduct a univariate sorting test. CBSAs are sorted into quintiles according to their CPRATIO values, and we calculate equal-weighted quintile averages for the alternative measure of gambling preference as well as the control variables. Table A2 summarizes the results. For the alternative measure, we find that gambling industry employment tends to increase with CPRATIO, and the difference between the highest and lowest quintiles is statistically significant. This suggests that employment in the gambling industry captures a construct that is consistent with CPRATIO[7].

For the control variables, employment in casino hotels do not show a substantial increase with CPRATIO. This outcome may stem from two factors. First, the presence of casino hotels is influenced not only by local gambling demand but also by tourist-oriented gambling activity. As a result, local gambling preference does not necessarily correlate with casino hotel employment or establishment density. Second, exceptions in gambling legality, such as tribal casinos, permit casino hotel operations in certain regions even when local gambling preference is low. In contrast, the number of nonemployer gambling establishments tends to increase with CPRATIO. This pattern suggests that such establishments complement minor local gambling demand and provide an additional proxy for community-level gambling activity.

Second, to further examine whether CPRATIO can explain variation in the alternative gambling preference measure and the additional control variables, we estimate a series of OLS regressions. Each dependent variable is regressed on CPRATIO and CBSA-level local control variables with state and year fixed effects. All continuous variables are winsorized at the 0.5 percent level and standardized prior to analysis. Table A3 presents the regression results. The gambling industry employment is positively and significantly associated with CPRATIO, indicating that higher CPRATIO values correspond to greater levels of the supply-side

[7]Note that the unconditional correlation between the CPRATIO and gambling industry employment is not large. The reason is that although Catholics believe gambling is not in itself contrary to justice, Catholic bishops' conferences have opposed the expansion of gambling industry due to its social harms. Therefore, in areas with high proportion of Catholic residents, even CPRATIO is high, the gambling industry may not high as well.

indicator. This supports the use of CPRATIO as a proxy for local gambling preference. In contrast, the per capita employment in casino hotels is negatively correlated with CPRATIO, reinforcing our decision to treat casino hotel variables as controls for tourist-oriented gambling demand rather than proxies for local gambling preference. Finally, for nonemployer gambling establishments, which serve as a complementary measure for minor local gambling activity, we also observe a significantly positive coefficient on CPRATIO. This suggests that higher CPRATIO values are associated with increased presence of nonemployer gambling businesses, further corroborating CPRATIO's relevance as an indicator of local gambling preference.

A.3 Additional Illustration

In this section, we address several concerns related to the use of alternative measure of local gambling preference.

First, several limitations prevent us from using the alternative measure as the basis for our primary analysis. Although employment and establishment data in the gambling industry may more directly reflect local gambling preferences, these indicators represent supply-side dynamics rather than demand-side behavior. Furthermore, they are influenced by factors such as local legality and economic conditions, which introduces additional noise into the measurement. Another key limitation is the restricted coverage. The alternative measure span only 460 CBSAs, encompassing approximately 1,187 counties, and cover around 1 million loans. In contrast, CPRATIO includes data for nearly 3,000 counties and covers almost the entire sample of 3 million loans. This broader coverage is especially critical when analyzing lender-specific effects and borrower preference heterogeneity.

Second, we rely on CBSA-level data rather than county-level data to construct our gambling preference measures. The rationale for using a broader geographic and economic unit is that residents in most U.S. metropolitan areas can easily commute within the city. Consequently, individuals residing in counties with few gambling establishments may work or gamble in adjacent counties with greater access. As a result, low employment or establishment counts in a given county do not necessarily reflect low gambling preference among its residents. By contrast, CBSAs are sufficiently large that cross-CBSA commuting for work or gambling is relatively rare. Instances of inter-CBSA gambling, such as through tourism, are mostly captured by the casino hotel measures. Therefore, the CBSA level offers a more robust and stable proxy for local gambling preferences than county-level data.

Third, we restrict our sample to CBSAs where establishments exist in all three categories: gambling industry, casino hotels, and nonemployer gambling businesses. This selection improves the clarity and reliability of our gambling preference measures. In such areas, false zeros, resulting from legal prohibitions, data suppression, or missing supply, are minimized. In untabulated regressions, when we replace missing values with zeros and include all CBSAs, the coefficient on the gambling preference measure remains positive but becomes only marginally significant. Furthermore, when we split the sample into high and low per capita establishment groups, the coefficient is strongly significant in the high-establishment group but insignificant in the low group. This pattern suggests that second-lien misrepresentation does rise with elevated local gambling preference, whereas low values in the measure may reflect extraneous noise rather than genuinely low gambling tendencies.

Table A1
Descriptive Statistics for Alternative Gambling Preference Measures

	N	Mean	SD	P10	P25	P50	P75	P90
Gambling Industries (NAICS 7132, CBSA-year level)								
Employment (per 10,000 residents)	1,201	19.60	42.74	0.22	0.69	2.53	16.41	59.50
Casino Hotel (NAICS 72110, CBSA-year level)								
Employment (per 10,000 residents)	258	118.49	264.87	0.07	0.50	16.90	77.41	389.73
Nonemployer Gambling (NAICS 7132, CBSA-year level)								
Establishment (per 10,000 residents)	1,239	0.38	0.50	0.00	0.14	0.27	0.45	0.76

The table presents descriptive statistics for the alternative local gambling preference measure and its related controls, covering the sample period from 2005 to 2007. We report the number of observations, mean, standard deviation, 25th percentile, 50th percentile, and 75th percentile.

Table A2
Univariate Sorting

Quintile	Gambling Industries Employment	Casino Hotel Employment	Nonemployer Gambling Establishment
Low	16.2	114.20	0.28
Q2	11.77	140.31	0.35
Q3	18.67	49.83	0.43
Q4	26.47	95.67	0.40
High	24.92	219.91	0.46
High-Low	8.66	105.72	0.18
t-Statistic	2.02	1.72	4.01

The table presents univariate sorting results for three gambling preference related measures: per capita employment in the gambling industry (Gambling Industries Employment), per capita employment in the casino hotel industry (Casino Hotel Employment), and per capita number of nonemployer gambling establishments (Nonemployer Gambling Establishment). CBSAs are sorted into quintiles based on CPRATIO, and for each quintile, the equal-weighted average of the five gambling preference related measures is computed. The table reports White robust t-statistics for the difference in mean values between the High and Low CPRATIO quintiles.

Table A3
Alternative measures estimate

	(1) Gambling Industry Employment	(2) Casino Hotel Employment	(3) Nonemployer Gambling Establishment
CPRATIO	0.107**	-0.166**	0.123***
	(0.042)	(0.083)	(0.038)
REL	-0.155***	-0.074	-0.164**
	(0.058)	(0.159)	(0.072)
Education	0.010	-0.816***	-0.098
	(0.055)	(0.247)	(0.070)
Married	0.072	-0.540***	0.001
	(0.045)	(0.167)	(0.058)
Income	-0.224***	1.084***	0.129
	(0.066)	(0.287)	(0.082)
Urban	0.081	-0.463***	-0.028
	(0.067)	(0.117)	(0.052)
Over65	0.083**	0.053	0.035
	(0.041)	(0.117)	(0.051)
Male-female ratio	0.061	-0.008	-0.038
	(0.043)	(0.084)	(0.036)
Minority	0.046	0.211*	-0.045
	(0.044)	(0.111)	(0.049)
Part I crime	-0.127***	-0.146	-0.020
	(0.040)	(0.129)	(0.037)
Part II crime	0.069*	0.061	-0.041
	(0.037)	(0.106)	(0.039)
Democratic	0.003	0.088	0.214***
	(0.062)	(0.143)	(0.075)
Total population	-0.063	-0.008	-0.024
	(0.048)	(0.126)	(0.062)
Unemployment	0.021	-0.116	-0.001
	(0.060)	(0.091)	(0.066)
HPA	0.011	0.123	0.088*
	(0.044)	(0.092)	(0.052)
State FE	Y	Y	Y
Half-year FE	Y	Y	Y
Observations	1,015	227	1,042
Adj. R^2	0.331	0.540	0.281

The table shows OLS estimates from regressions where the dependent variables are the alternative gambling preference measure and related control variables. Religiosity and the geographic controls are aggregated to CBSA level. All continuous variables are winsorized at the 0.5 percent level and then standardized. White robust standard errors are reported in parentheses. $*p < 0.10$, $**p < 0.05$, and $***p < 0.01$.

B Diff-in-disc concerns

To validate our application of the diff-in-disc approach in comparing the number of loans and default rates between high and low gambling preference areas, we address the following concerns in this section.

B.1 Assumptions for Applying Diff-in-disc

According to Grembi et al. (2016), the validity of the diff-in-disc approach depends on three assumptions. First, all potential outcomes must be continuous around the cutoff. In our setting, this means no manipulation of FICO scores around the cutoff point of 620. Since the data on the distribution of FICO scores in the U.S. population is not available to us, we argue this assumption by referencing the findings in Keys et al. (2010). They found that the distribution of FICO scores across the population is smooth using data from an anonymous credit bureau. Additionally, by exploring the reversal of anti-predatory lending laws in Georgia and New Jersey, they found that borrowers were either unaware of the differential screening around the threshold or unable to quickly manipulate their FICO scores[8].

Second, in the absence of treatment, the effect of the confounding policy around the cutoff is constant over the "diff" part. To fulfill this assumption, we check whether the pattern holds for full documentation loans around FICO 620, which have no treatment but similar confounding factors. Table B1 shows the results for the number of loans (panel A) and default rate (panel B) estimated using full-documentation loans. For full-documentation loans, the number of loans shows a minimal jump, and the difference between high and low gambling preference areas is slight. This evidence indicates that without the ease of securitization, having a FICO score below or above 620 would not cause the number of loans to increase differently between high and low gambling preference areas. Additionally, there is no jump in the default rate for all types of loans, and the difference in the increase of the default rate between high and low gambling preference areas is minor. These findings illustrate that without the ease of securitization, lenders do not have lax screening standards for any type of loan and do not screen differently between high and low gambling preference areas.

Third, the effect of the treatment around the cutoff does not depend on the confounding policy. This assumption states that there should be no interaction between the treatment and the confounding policy. In other words, the pre-determined outcomes and covariates should have similar jumps (or no jumps) between high and low gambling preference areas. We test this assumption by estimating equation 6 with pre-determined outcomes and covariates as the

[8] According to the rating agency (Fair Isaac), strategic manipulation of FICO scores is difficult.

dependent variable and also add other control variables in the regression. We choose the same bandwidth (10) and kernel (uniform) as the main tests. The results are reported in Table B2. Column (1) presents the results of loan characteristics, showing that most characteristics have similar jumps between high and low gambling preference areas. The proportions of negative amortization and second-home loans are significant, but we argue that this does not affect our conclusion for several reasons. First, although the absolute jump magnitude is different, the ratio of 620^+ to 620^- is close, i.e., negative amortization jumps from 6.74/2.97 percent to 15.82/6.61 percent (2.35/2.23 times) in high/low gambling preference areas. Second, the proportion of these specific loans is too low to have a meaningful effect on the number of loans with misrepresentation and the default rate (i.e., in this local range, second homes make up only 1.2 percent of all loans and only 0.6 percent of misreported loans). Column (2) reports the results of borrower characteristics, showing that most characteristics have similar jumps between high and low gambling preference areas. The ratios of older people, the male-female ratio, and the Part II crime rate are significant, but their influence is minor because the magnitude is too small, i.e., the average proportion of older people, the male-female ratio, and the Part II crime rate in the local range are 12.2 percent, 28.89 percent, and 330 per 10,000 residents, while the differences are only 0.27 percent, 0.003 percent, and -9.5 per 10,000 residents, respectively.

B.2 Sensitivity to Regression Choices

Beyond these three assumptions, we also check whether the findings are sensitive to the inclusion of control variables, the choice of bandwidth, and the choice of estimation kernel. We estimate models with and without covariates, choosing bandwidths of 8, 10, and 12, and using uniform or triangular kernels. Table B3 reports the ratio of estimated values at FICO 620^+ to 620^- for the number of loans[9]. The magnitude of jumps is similar to the baseline results: the number of different types of loans increases significantly due to the ease of securitization, and the jump for loans with misreported seconds is smaller in high gambling preference areas than in low gambling preference areas. These results confirm the robustness of the findings that lenders do not play a larger role in increasing second-lien misrepresentation in high gambling preference areas. Table B4 shows that the default rate of loans with misreported seconds is small and insignificant in all cases. Moreover, although not significant, the jump in high gambling preference areas is smaller than the jump in low gambling preference areas in most cases. These robustness tests show that lenders in high gambling preference areas are not more inclined to increase such misrepresentation than

[9] Since the number of loans is counted at the level of gambling preference areas while control variables are at the level of counties, control variables could not be included in the tests.

lenders in low gambling preference areas.

B.3 Placebo Test

Lastly, we conduct a placebo test to evaluate the possibility that our results are driven by chance. Specifically, following Goodman and Mayer (2018), we implement the diff-in-disc estimations at false FICO score thresholds below and above 620. Our inferences on the lender's role come from two findings from the diff-in-disc estimations: (1) the number of loans with misreported seconds increases much less in high gambling preference areas at FICO 620, and (2) the default rate of loans with misreported seconds either shows a small and insignificant jump in high gambling preference areas or a jump that is smaller in high gambling preference areas than in low gambling preference areas. Thus, at these false thresholds, we expect to find (1) no systematic difference in the increase of the number of loans with misreported seconds between high and low gambling preference areas as in our baseline results, and (2) no situation where the default rate of loans with misreported seconds shows a meaningful upward jump in high gambling preference areas and a jump that is larger in high gambling preference areas than in low gambling preference areas. To maintain a similar setting and avoid potential jumps in the misrepresentation rate found in the literature, we conduct the tests within the range of 601 to 639[10]. For the number of loans, we exclude the range of 615 to 625 to stay sufficiently away from the true threshold that could cause noise in jump estimation. Figure B1 plots the cumulative density function of the 28 placebo point estimates for the number of loans with misreported seconds. All of the placebo coefficients are greater than our estimated coefficient (-2.063) and smaller than the absolute value of our estimated coefficient with a small magnitude. We also perform a t-test on the placebo coefficients to check the null hypothesis that the true mean equals 0, and the p-value of this test is 0.819, which fails to reject the null hypothesis. These placebo tests support the idea that the difference in discontinuities for the number of loans with misreported seconds between high and low gambling preference areas is not due to chance. For the default rate, we only exclude the true threshold since no jump exists at the true threshold, so estimations at the scores near it would not be affected. Figure B2 plots the cumulative density function of the placebo point estimates for the default rate of loans with misreported seconds. Most placebo coefficients are smaller than the absolute value of our estimated coefficient (0.052). The larger ones are also insignificant, and they show an insignificant small jump in high gambling preference areas. The t-test on the placebo coefficients that checks whether the true mean equals 0 provides a p-value of 0.638. These placebo tests support the idea that

[10] From Figure 3 in Griffin and Maturana (2016) about the misrepresentation rate, we can see clear jumps at FICO 600 and 640.

lenders did not have laxer screening in high gambling preference areas.

Table B1
Discontinuities of Full-documentation Loans Around the Credit Threshold

	(1) High	(2) Low	(3) All	(4) High-Low
Panel A. Number of loans (Est. 620+/620-)				
All loan	1.244	1.154	1.186	0.090
				(2.27)
Reported second	1.183	1.152	1.161	0.031
				(0.58)
Misreported second	1.325	1.216	1.251	0.109
				(1.03)
Panel B. Default rate				
All loan	0.002	0.008	0.007	-0.007
	(0.25)	(0.07)	(0.35)	(-0.54)
Reported second	-0.023	0.001	-0.008	-0.024
	(-0.93)	(0.04)	(-0.39)	(-0.72)
Misreported second	-0.014	0.030	0.016	-0.043
	(-0.87)	(0.03)	(-0.47)	(-1.05)

The table presents the results of the estimations in Table 7 and Table 8 using full-documentation loans. Panel A reports the estimates for the number of loans at each FICO score. Panel B reports the estimates for loans that become 90 days or more delinquent using the MBA method in the first three years after origination. Using local linear regressions of the RDD approach, we estimate the number of loans and the default rate (with control variables) at FICO 620^{-} and FICO 620^{+} and compute the ratio of the estimated number of loans at FICO 620^{+} to 620^{-}. Using the diff-in-disc approach, we estimate the difference between high and low gambling preference areas. We perform the estimation for all loans, loans with correctly reported simultaneous seconds, and loans with misreported simultaneous seconds. The results for loans in high gambling preference areas, loans in low gambling preference areas, loans in all areas, and the difference between high and low gambling preference areas are reported in columns (1) to (4), respectively. For discontinuities, bias-corrected t-values (standard errors clustered at the county level) following Calonico et al. (2014) are reported in parentheses. For the difference in the ratio of the number of loans/default rate between high and low gambling preference areas, t-statistics based on heteroskedasticity-consistent standard errors/standard errors clustered at the county level are reported in parentheses.

Table B2
Estimates of the discontinuities in observable covariates

	(1) loan characteristics		(2) local characteristics
Interest rate (%)	-0.005	REL	-0.004
	(0.041)		(0.003)
Balance	0.015	Education (%)	-0.343
	(0.012)		(0.200)
LTV (%)	0.169	Married (%)	-0.113
	(0.367)		(0.089)
ARM	-0.010	Income	0.005
	(0.013)		(0.007)
Negative amortization	0.025***	Urban (%)	0.563
	(0.007)		(0.331)
Option ARM	0.001	Over65 (%)	0.271**
	(0.002)		(0.108)
Prepayment penalty	0.004	Male-female ratio	0.003**
	(0.015)		(0.001)
Cash-out	0.003	Minority (%)	-0.430
	(0.013)		(0.307)
No-cash-out	0.003	Part I crime	-3.718
	(0.007)		(4.896)
Investment	0.007	Part II crime	-9.473*
	(0.007)		(5.703)
Second-home	0.007**	Democratic	-0.003
	(0.003)		(0.014)
		Total population (ln)	0.017
			(0.025)
		Unemployment (%)	0.063
			(0.058)
		HPA	-0.005
			(0.004)

The table reports the results of diff-in-disc regression for pre-determined outcomes and covariates using Equation 6 with other controls. The dependent variables are the pre-determined outcomes and covariates. Column (1) shows the tests for loan characteristics, and column (2) shows the tests for borrower characteristics. As in the main test, the bandwidth is set to 10 and the kernel is uniform. Standard errors clustered at the county level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table B3
Sensitivity of Results for Number of Loans

	Uniform				Triangular			
	High	Low	All	H-L	High	Low	All	H-L
Bandwidth = 8								
All loan	2.230	2.242	2.237	-0.013	2.250	2.204	2.225	0.046
				(-0.13)				(0.44)
Reported second	5.469	4.693	4.999	0.776	5.761	5.235	5.446	0.526
				(0.96)				(0.56)
Misreported second	2.709	5.095	3.957	-2.385	2.496	5.128	3.826	-2.632
				(-2.85)				(-3.21)
Bandwidth = 10								
All loan	2.240	2.289	2.266	-0.049	2.240	2.234	2.237	0.006
				(-0.52)				(0.06)
Reported second	5.498	4.581	4.939	0.917	5.616	4.971	5.228	0.645
				(1.23)				(0.74)
Misreported second	2.763	4.826	3.838	-2.063	2.609	5.133	3.900	-2.524
				(-2.83)				(-3.20)
Bandwidth = 12								
All loan	2.199	2.245	2.224	-0.047	2.237	2.262	2.250	-0.025
				(-0.50)				(-0.27)
Reported second	5.209	4.602	4.846	0.607	5.520	4.845	5.114	0.676
				(0.91)				(0.84)
Misreported second	2.656	4.388	3.590	-1.732	2.668	4.971	3.858	-2.302
				(-2.60)				(-3.20)
Bandwidth = 14								
All loan	2.197	2.199	2.198	-0.002	2.222	2.257	2.241	-0.036
				(-0.02)				(-0.39)
Reported second	5.278	4.666	4.915	0.612	5.424	4.788	5.043	0.636
				(0.95)				(0.84)
Misreported second	2.515	3.940	3.275	-1.426	2.645	4.755	3.747	-2.110
				(-1.99)				(-3.18)

The table reports the sensitivity of RDD and diff-in-disc regression for the number of loans to the choice of bandwidth and estimation kernel. For the difference between high and low gambling preference areas, t-statistics based on heteroskedasticity-consistent standard errors are reported in parentheses.

Table B4
Sensitivity of Results for Default Rate of Loans with Misreported Second

	Uniform				Triangular			
	High	Low	All	H-L	High	Low	All	H-L
Bandwidth = 8								
No control	0.005	0.019	-0.004	-0.014	-0.007	-0.016	-0.031	0.009
	(-0.24)	(-0.83)	(-1.18)	(-0.23)	(0.81)	(-1.43)	(-0.89)	(0.13)
With control	0.026	0.048	0.033	-0.022	-0.001	-0.012	-0.007	0.010
	(-0.39)	(-1.27)	(-1.27)	(-0.43)	(0.45)	(-1.73)	(-0.97)	(0.16)
Bandwidth = 10								
No control	0.001	0.052	0.014	-0.052	-0.004	0.011	-0.014	-0.015
	(-0.07)	(-0.86)	(-1.15)	(-0.98)	(0.32)	(-1.15)	(-1.07)	(-0.25)
With control	0.013	0.078	0.045	-0.065	0.011	0.025	0.016	-0.014
	(0.16)	(-0.92)	(-0.73)	(-1.41)	(0.06)	(-1.43)	(-1.06)	(-0.26)
Bandwidth = 12								
No control	-0.022	0.065	0.014	-0.087	-0.003	0.033	0.000	-0.036
	(0.30)	(-0.30)	(-0.50)	(-1.71)	(0.09)	(-1.00)	(-1.14)	(-0.65)
With control	-0.004	0.082	0.042	-0.085	0.011	0.050	0.030	-0.038
	(0.33)	(0.07)	(0.15)	(-1.95)	(0.12)	(-1.12)	(-0.82)	(-0.80)
Bandwidth = 14								
No control	0.007	0.071	0.033	-0.064	-0.007	0.046	0.007	-0.053
	(-0.47)	(0.11)	(-0.72)	(-1.39)	(0.06)	(-0.59)	(-0.88)	(-1.03)
With control	0.017	0.081	0.051	-0.064	0.008	0.060	0.035	-0.052
	(-0.29)	(0.79)	(0.26)	(-1.53)	(0.13)	(-0.43)	(-0.35)	(-1.18)

The table reports the sensitivity of RDD and diff-in-disc regression for the default rate of loans with misreported seconds to the inclusion of control variables, the choice of bandwidth, and the choice of estimation kernel. For discontinuities, bias-corrected t-values (standard errors clustered at the county level) following Calonico et al. (2014) are reported in parentheses. For the difference between high and low gambling preference areas, t-statistics based on heteroskedasticity-consistent standard errors are reported in parentheses.

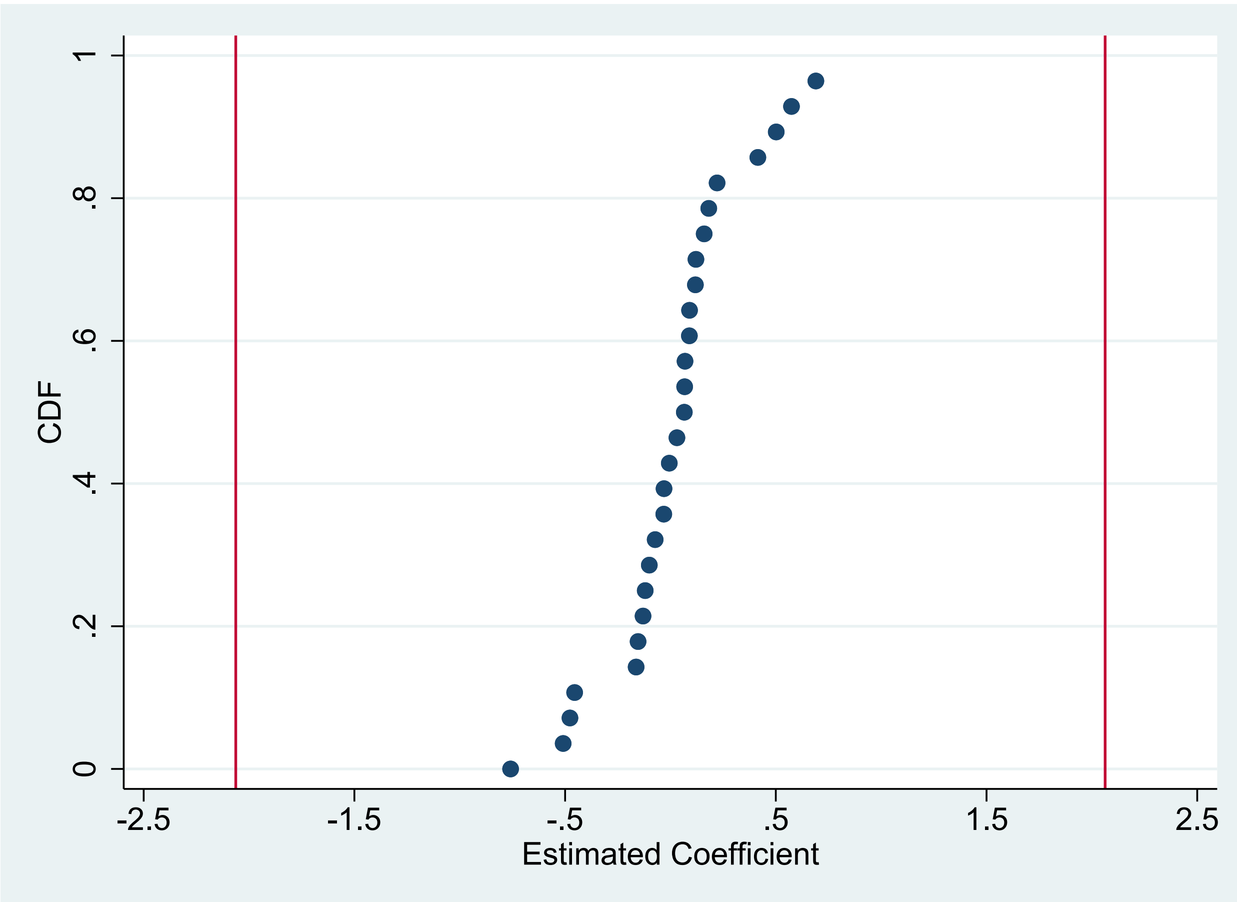

Figure B1
Placebo Tests for Number of Loans with Misreported Second
The figure plots the empirical c.d.f. of the estimated coefficient for the number of loans with misreported seconds from a set of diff-in-disc estimations at false FICO score thresholds below and above 620 (i.e., any score from 601 to 614 and any score from 626 to 639). The vertical lines show the benchmark estimate (-2.063) and its positive value (2.063).

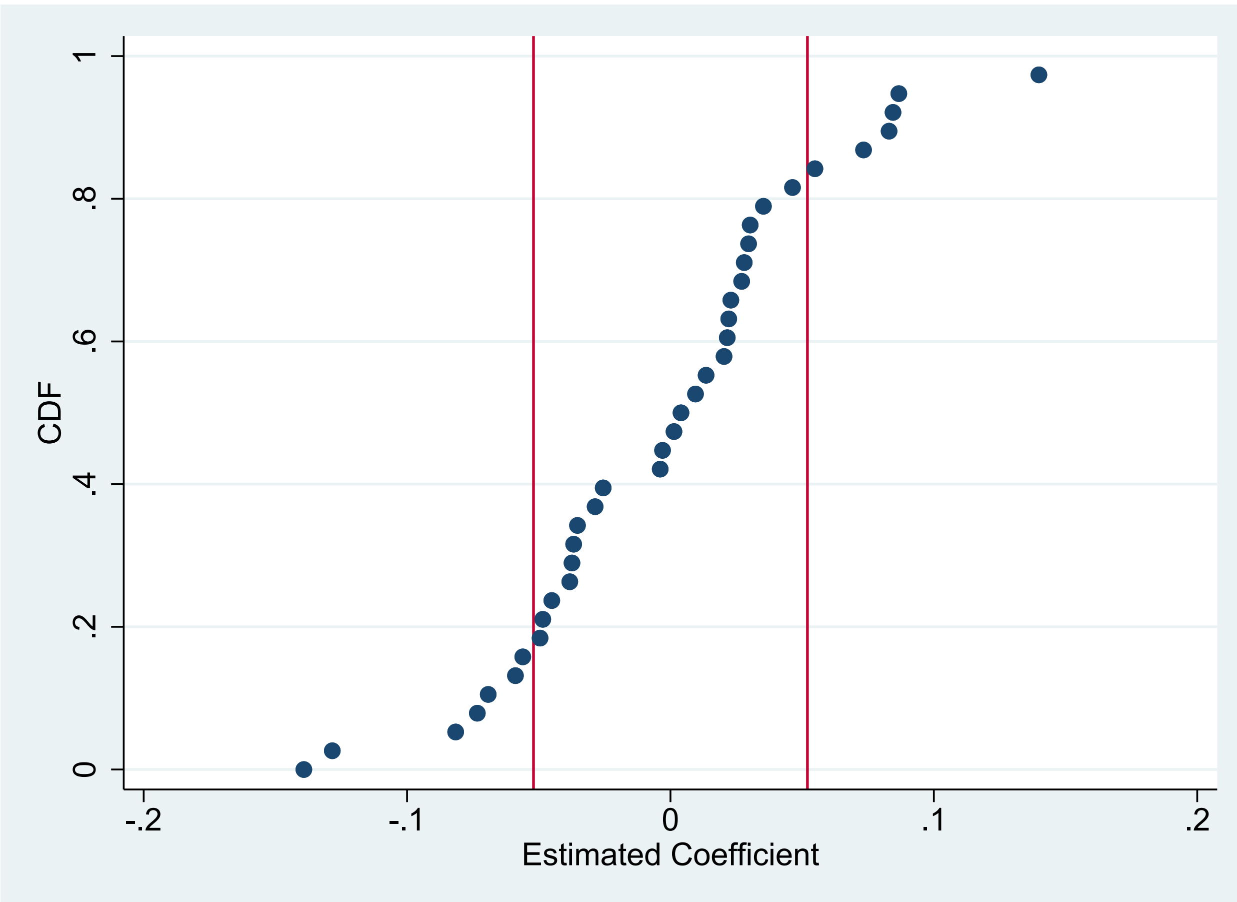

Figure B2
Placebo Tests for Default Rate of Loans with Misreported Second
The figure plots the empirical c.d.f. of the estimated coefficient for the default rate of loans with misreported seconds from a set of diff-in-disc estimations at false FICO score thresholds below and above 620. The vertical lines show the benchmark estimate (-0.052) and its positive value (0.052).